

Deciphering deep-sea chemosynthetic symbiosis by single-nucleus RNA-sequencing

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Abstract

Bathymodioline mussels dominate deep-sea methane seep and hydrothermal vent habitats and obtain nutrients and energy primarily through chemosynthetic endosymbiotic bacteria in the bacteriocytes of their gill. However, the molecular mechanisms that orchestrate mussel host-symbiont interactions remain unclear. Here, we constructed a comprehensive cell atlas of the gill in the mussel *Gigantidas platifrons* from the South China Sea methane seeps (1100m depth) using single-nucleus RNA sequencing (snRNA-seq) and whole-mount *in situ* hybridisation. We identified 13 types of cells, including three previously unknown ones, and uncovered unknown tissue heterogeneity. Every cell type has a designated function in supporting the gill's structure and function, creating an optimal environment for chemosynthesis, and effectively acquiring nutrients from the endosymbiotic bacteria. Analysis of snRNA-seq of *in situ* transplanted mussels clearly showed the shifts in cell state in response to environmental oscillations. Our findings provide insight into principles of host-symbiont interaction and the bivalves' environmental adaption mechanisms.

eLife assessment

This study provides an **important** cell atlas of the gill of the mussel *Gigantidas platifrons* using a single nucleus RNA-seq dataset, a resource for the community of scientists studying deep sea physiology and metabolism and intracellular host-symbiont relationships. The evidence supporting the conclusions is **convincing** with high-quality single-nucleus RNA-sequencing and transplant experiments. This work will be of broad relevance for scientists interested in host-symbiont relationships across ecosystems.

Introduction

Mutualistic interactions between multicellular animals and their microbiota play a fundamental role in animals' adaptation, ecology, and evolution^{1,2}. By associating with symbionts, host animals benefit from the metabolic capabilities of their symbionts and gain fitness advantages that allow them to thrive in habitats they could not live in on their own^{1,3}. Prime examples of such symbioses are bathymodioline mussels and gammaproteobacteria endosymbionts⁴. The bathymodioline mussels occur worldwide at deep-sea chemosynthetic ecosystems, such as cold-seeps, hydrothermal vents, and whale falls⁴. The host mussel acquired sulfur-oxidising (SOX) and/or methane-oxidising (MOX) symbiont through horizontal transmission at their early life stage³. Their symbionts, which are hosted in a specialised gill epithelial cell, namely, the bacteriocytes^{5,6}, utilise the chemical energy from the reduced chemical compounds such as CH₄ and H₂S released from cold seeps or hydrothermal vents to fix carbon and turn into carbon source for the host mussel^{7,8}. The ecological success of the bathymodioline symbioses is apparent: the bathymodioline mussels are often among the most dominant species in the deep-sea chemosynthetic ecosystems^{9,10}. Thus, it is critical to know how the bathymodioline mussels interact with the symbionts and maintain the stability and efficiency of the symbiosis.

The gill structure of bathymodioline mussels has undergone remarkable adaptations at the molecular, cellular, and tissue levels to support their deep-sea chemosynthetic lifestyle^{11,12}. Compared to shallow water mussels¹³, the gill filaments of bathymodioline mussels have enlarged surfaces allowing them to hold more symbionts per filament (Supplementary Fig. S1). This adaption requires not only novel cellular and molecular mechanisms to maintain and support the enlarged gill filament structure, but also a strong ciliary ventilation system to pump vent/seep fluid to fuel the symbiotic bacteria. Previous studies based on whole-genome sequencing and bulk RNA-seq projects have shown that genes in the categories of nutrient transporters, lysosomal proteins, and immune receptors are either expanded in the host mussel's genome or up-regulated in the gill^{12, 14–17}, suggesting that these genes involved in host-symbiont interaction. Though providing deep molecular insights, these studies mainly used homogenised tissues that average genes' expression levels amongst different cell types and eliminate cell and gene spatial distribution information^{18–21}. In addition, the broad expression and function of potentially 'symbiosis-related' proteins also greatly limited data interpretation. Therefore, a systemic atlas of gill cell types and the descriptions of cell-type-specific gene expressional profiles are warranted to a better understanding of the host-symbiont interaction and environmental adaptation mechanisms of the bathymodioline symbiosis.

In recent years, single cell/nucleus RNA-sequencing (sc/sn RNA-seq) technologies have become one of the preferred methods for investigating the composition of complex tissue at the transcriptional level²². snRNA-seq has several advantages, such as compatibility with frozen samples, elimination of dissociation-induced transcriptional stress responses, and reduced dissociation bias²³. These advantages are significant for deep-sea and other ecological studies because cell and molecular biology facilities are not available in the field. To examine the cellular and molecular mechanisms of the environmental adaptations and host-symbiont interactions in bathymodioline mussels, we conducted a single nucleus RNA-sequencing-based transcriptomic study. We analysed the gill symbiosis in the dominate deep-sea mussels inhabiting the F-site cold-seep *Gigantidas platifrons*²⁴, which hosts a single a MOX endosymbiont population that consist of only one 16S rRNA phylotype. Our work provides a proof-of-principle for environmental adaptation mechanisms study in non-traditional but ecologically important organisms with single-nucleus RNA-sequencing technologies.

Results and discussion

1. *G. platifrons* deep-sea *in situ* transplant experiment and single-cell transcriptomic sequencing

We conducted a *G. platifrons* deep-sea *in situ* transplant experiment at the “F-Site” Cold-seep (~1117 m depth) and retrieved three groups of samples (**Fig. 1A**), as follows: the ‘Fanmao’ (meaning prosperous) group, which comprised the mussels collected from methane-rich (~40 μM) site of the cold-seep, where the animals thrived; the ‘starvation’ group, which comprised the mussels collected from the methane-rich site and then moved to a methane-limited (~0.054 μM) starvation site for 14 days before retrieval; and the ‘reconstitution’ group, which comprised the methane-rich site mussels moved to the ‘Starvation’ site for 11 days and then moved back to the methane-rich site for another 3 days.

We next performed single nucleus RNA-sequencing (snRNA-seq) of the gill posterior tip of the three groups of samples using the microwell-based BD Rhapsody platform™ Single-Cell Analysis System (**Fig. 1A**). After quality control, we obtained 9717 (Fanmao), 21614 (Starvation), and 28928 (Reconstitution) high-quality single nuclei transcriptomic data.

2. Cell atlas of *G. platifrons* gill

To unravel the intricate cellular composition, we utilised a reciprocal PCA (RPCA) strategy to project cells in the three samples onto the same space based on conserved expressed genes among them²⁵. This strategy maximises the number of cells per cluster regardless of the organism’s state and therefore maximises genes per cluster²⁶.

Through the implementation of Seurat, we revealed 14 cell clusters, each was associated with a set of marker genes (**Figs. 1B-D**)^{27,28}. Given the limited availability of canonical marker genes for *G. platifrons* and molluscs in general, we undertook a meticulous approach to characterize each cell cluster. This involved: 1) examining the cluster marker genes’ functions, 2) identifying the expression pattern of cluster marker gene using whole-mount *in situ* hybridisation (WISH) and/or double-fluorescent *in situ* hybridisation (FISH) analyses, and 3) conducting scanning electron microscopy (SEM) and transmission electron microscopy (TEM) analyses. We successfully identified and characterized 13 of the 14 cell clusters, which could be categorized into four major groups, including 1) the supportive cells, 2) the ciliary cells, 3) the proliferation cells, and 4) the bacteriocytes. One cell cluster remained ambiguous. To assess the robustness of each cell cluster, we employed a bootstrap sampling and clustering algorithm examining the similarity among clusters and obtained strong support for all clusters through a combined analysis of the three samples (Supplementary Fig. S2A). Furthermore, when examining each sample individually, we found that the majority of clusters demonstrated robust support, with the exception of the three ciliary cell clusters, which showed overlaps of assignment probabilities among them (Supplementary Fig. S2 B-D). These three cell types are derived from a same precursor, and exhibited relatively lower numbers of nuclei, resulting in a reduced availability of genes per cell type, which is particularly true for the food grove ciliary cell (Supplementary Table S1). By integrating the spatial and functional annotations of these cell clusters, we gained insight into their collective efforts in supporting the symbiotic relationship and maintaining the high-efficiency chemosynthetic system within the gill tissue.

Supportive cells

We have identified four supportive cell types, namely, inter lamina cells basal membrane cells (BMC)1, BMC2, and mucus cells (**Figs 2A, B** and **C**).

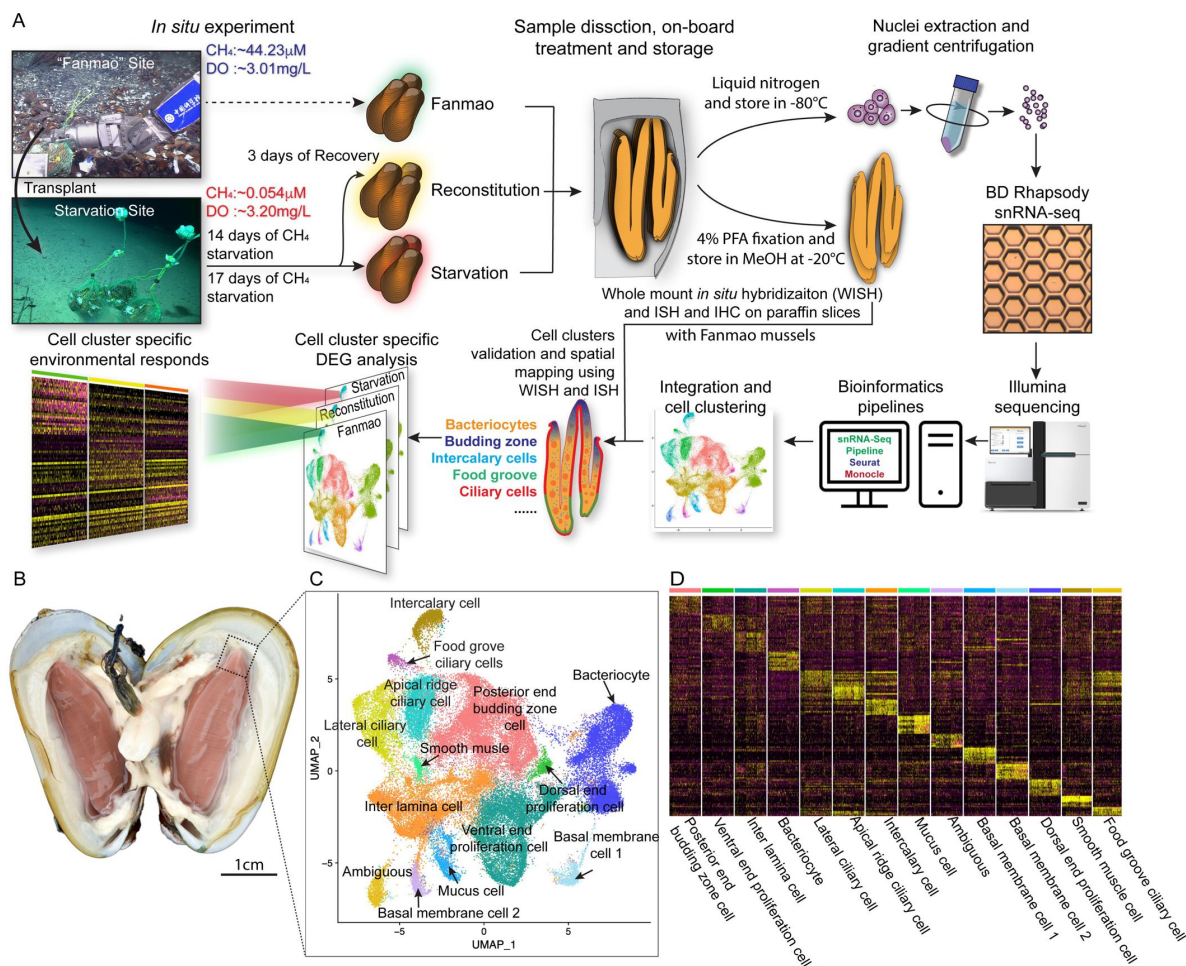


Figure 1

Identification of 14 cell types in the gill of deep-sea symbiotic mussel *Gigantidas platifrons*. A: Overall experimental scheme describing the deep-sea *in situ* transplant experiment, the sample preparation procedures and the single-cell analysis and validation pipeline. Three *G. platifrons* samples were included in the present study: 'Fanmao,' Starvation and Reconstitution. The cell nucleus was extracted from each sample, which included a pool of gill posterior tip of three mussels. The snRNA-seq libraries were constructed according to the BD Rhapsody single nuclei 3. protocol. Cell population-specific markers were validated by WISH and ISH. B: The image shows the posterior end of the gill of *G. platifrons*. C: UMAP representation of *G. platifrons* gill single cells. Cell clusters are coloured and distinctively labelled. D: Heat map profile of markers in each cluster. The colour gradient represents the expression level of every single cell.

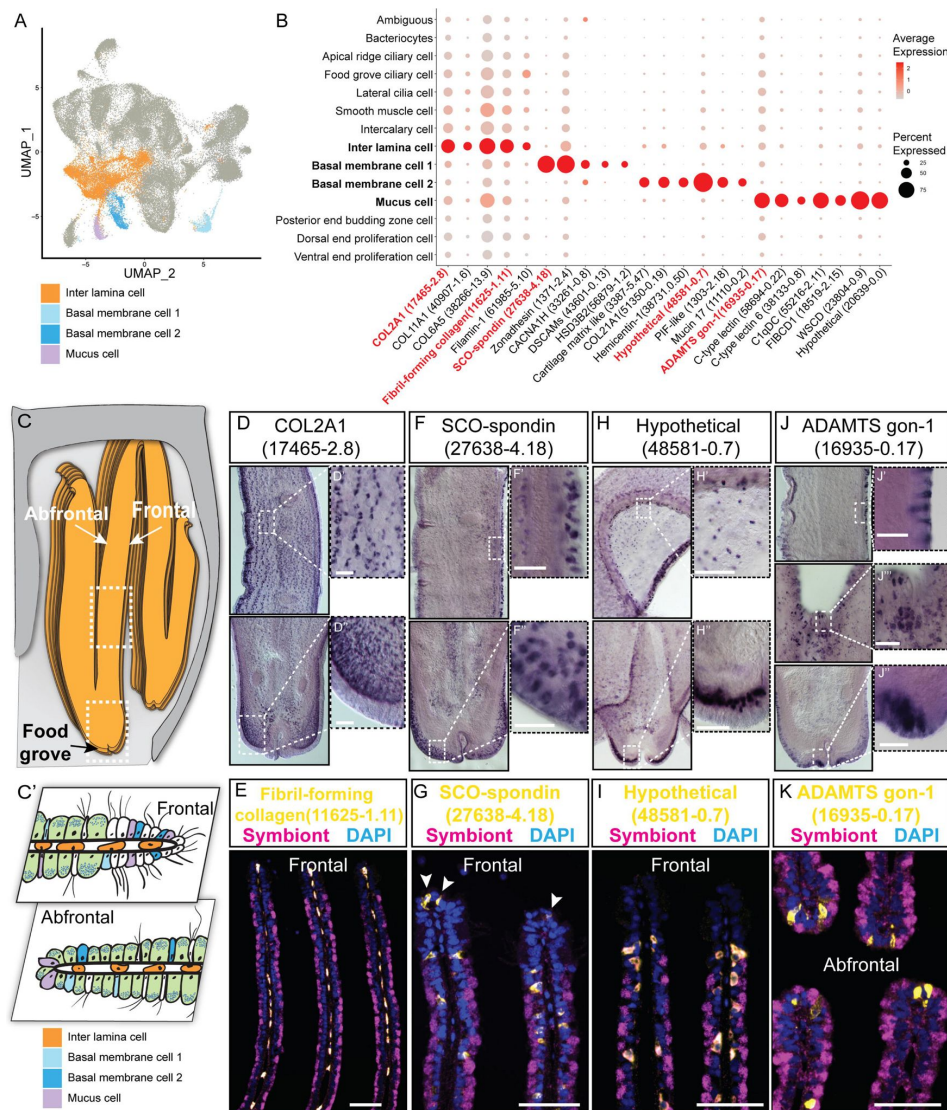


Figure 2

Supportive cell populations of *G. platifrons* gill. **A**: UMAP representation of the four supportive cell populations. **B**: Expression profiles of the cell markers that are specific or enriched in the supportive cell populations. The sizes of the circles represent the percentages of cells in those clusters that expressed a specific gene. Genes shown in red were validated by WISH or double FISH. **C** and **C'**: Schematics demonstrating the overall structural (panel **C**) and supportive cell distribution (**C'**). **D**, **F**, **H**, and **J**: WISH characterisation of the selected representative cell population markers. **E**, **G**, **I**, and **K**: Double FISH characterisation of the selected representative cell population markers. The white arrowheads in **G** indicate the BMC1 cells located at the outer rim of gill slice. Scale bar: 50µm.

The inter lamina cells, marked by high expression of fibrillar-forming collagens, were the cells located between the two layers of basal membranes (**Fig 2D and E**). These cells were previously identified as amoeboid cells¹³, and their function has not been explored. The inter lamina cells were densely distributed around the food groove and at the rim of the gill filament. While on the middle part of the gill filament, the inter lamina cells distributed in parallel rows. It might help to connect the two sheets of basal membranes and maintain the spatial integrity of the gill filament. In addition, we observed the enrichment of ribosomal proteins involved in protein synthesis (Supplementary Table S2), indicating a high metabolic rate²⁹ of inter lamina cells. This implies that the inter lamina cells may process the nutrients acquired from the symbiont.

We have also identified two types of BMC populations which have not been recognised before. The BMC1 (**Fig. 2F and G**) expressed genes encoding extracellular matrix and adhesive proteins (**Fig. 2B** and Supplementary Table S2), such as SCO-spondin (Bpl_scaf_27638-4.18), Zonadhesin (Bpl_scaf_1371-2.4), Cartilage matrix proteins (Bpl_scaf_16371-0.14 and Bpl_scaf_3387-5.47), and Collagen alpha-1(XI) chain protein (Bpl_scaf_51350-0.19). BMC2 (**Fig. 2H and I**) was distinctive from BMC1 in the context of expression (**Fig 2A**) and marker genes (**Fig 2B**). Genes encoding hemicytins (Bpl_scaf_38731-0.50 and Bpl_scaf_7293-0.14), which are known to stabilise the cells' contact with the basal membrane³⁰, were highly expressed in BMC2. Thus, these two cell types are likely to help building the basal lamina, and stabilising the epithelial-derived cells, such as bacteriocytes and intercalary cells, on the surface of the basal membrane.

Mucus cells are specialised secretory cells with intracellular mucus vacuoles¹³. In shallow-water filter-feeding bivalves, mucus cells secrete mucus, which cooperates with the ciliary ventilation system to capture, process, and transport food particles to their mouths, known as filter feeding mechanism^{31–34}. Deep-sea mussels may not necessarily retain this function since they do not normally acquire food resources such as phytoplankton and planktonic bacteria but obtain most of nutrient through their chemosynthetic symbiont. Thus, it was hypothesised that mucus cells are involved in other biological functions such as immune responses to pathogens¹⁸. Herein, genes encoding proteins with microbe-binding functionalities were enriched in mucus cells (**Fig. 2B** and Supplementary Table S2), such as C-type lectins (Bpl_scaf_58694-0.22 and Bpl_scaf_38133-0.8), C1q domain-containing protein (Bpl_scaf_55216-2.11), and Fibrinogen C domain-containing protein (Bpl_scaf_18519-2.15)^{35,36}. The expression of similar immune genes was upregulated in a shallow water mussel *Mytilus galloprovincialis* when challenged by a pathogenic bacteria³⁷. Our WISH and double-FISH analyses showed that mucus cells were embedded within the outer rim cilia and scattered on the gill lamella alongside the bacteriocytes (**Figs. 2J and 2K**). These data collectively suggest that mucus cells may help mussel maintain the immune homeostasis of gill. Interestingly, WISH and double-FISH analyses showed that mucus cells were also distributed alongside the gill lamella's food groove and the inner edge, where the density of bacteriocytes is low (**Fig. 2J**). Because the food groove is the main entry of food practical to the labial palps and mouth³⁸, this distribution pattern implies that mucus cells may be also involved in capturing planktonic bacteria and sending them to the mouth.

Ciliary and smooth muscle cells

A remarkable feature of bathymodioline mussel's gill is its ciliary ventilation system, which constantly agitates the water and provides the symbiont with the necessary gas³⁹. We identified four types of ciliary cells (**Figs. 3A and 3B**): apical ridge ciliary cells (ARCCs), food groove ciliary cells (FGCCs), lateral ciliary cells (LCCs), and intercalary cells (ICs), as well as a newly identified type of smooth muscle cells. All ciliary cells were marked by canonical cilium genes, such as genes encoding flagellar proteins, ciliary motor kinesin proteins, ciliary dynein proteins, and ciliary microtubules and were clearly distinguishable by specifically expressed genes (Supplementary Table S3 and Supplementary Fig. S3). ARCCs were characterized by expression of Tubulin alpha-1A chain Bpl_3489-0.37 (**Fig. 3C**) and Bpl_scaf_20631-1.16, which encodes homeobox Dlx6a-like protein (**Fig. 3D**) which is a marker of the apical ectodermal ridge⁴⁰.

FGCCs, which could be laded by expression of marker gene *Bpl_scaf_5544-0.0* (Fig. 3E), expressed genes encoding primary cilia development regulator Tubby-related protein 3⁴¹ (*Bpl_scaf_24834-2.3*) and primary ciliary cell structural protein TOG array regulator of axonemal microtubules protein 1^{42,43} (*Bpl_scaf_55620-1.3*), suggesting the FGCCs are the sensory ciliary cells that gather information from the surrounding environment. Both ARCCs and FGCCs were located around the ventral tip of the gill filament (Fig. 3F). LCCs were distributed as two parallel rows along the gill lamella's outer rim and ciliary disks' outer rim (Figs. 3G and 3H) and had highly expressed genes involved in cilium structure, cilium movement and ATP synthases (Fig. 3B and Supplementary Table S3), indicating that LCCs may have a strong ability to beat their cilia. Interestingly, we identified a group of previously unreported smooth muscle cells (SMCs) co-localised with the LCCs as showed by WISH (Fig. 3I and Supplementary Fig. S4). SMCs strongly expressed several Low-density lipoprotein receptors (LDL receptor) and LDLR-associated proteins⁴⁴ (Fig. 3B). In addition, these cells also expressed the angiotensin-converting enzyme-like protein⁴⁵ (*Bpl_scaf_29477-5.10*) and the 'molecular spring' titin-like protein⁴⁶ (*Bpl_scaf_56354-7.8*) (Supplementary Table S3). The expression of these genes could be commonly found in human vascular smooth muscle cells⁴⁷⁻⁴⁹. Collectively, we suspect that SMCs are involved in lateral cilium movement and the gill slice contraction.

ICs are the specialised ciliary cells surrounding the bacteriocytes (Figs. 3B and 3J). WISH analysis showed that the expressions of IC markers had apparent spatial variations (Fig. 3J). The expression of ICs marker rootletin⁵⁰⁻⁵² (*Bpl_scaf_22362-6.9*) was considerably higher in the ICs close to the frontal edge of the gill filament, and the expression gradually decreased along with the direction of inter lamina water flow (Fig. 3K and Supplementary Fig. S5), implying that ICs ventilate the water flow and the mucus through the gill filaments. Furthermore, compared with the other three types of ciliary cells, the ICs expressed several genes encoding transcription factors involved in determining cell fate (Fig. 3B and Supplementary Table S3), such as transcription factors Sox 8 (*Bpl_scaf_40595-5.30*) and Wnt pathway cell polarity regulator secreted frizzled-related protein 5 (*Bpl_scaf_57424-6.1*)⁵³, suggesting that the ICs might also play regulatory roles⁵⁴⁻⁵⁶.

Proliferation cells

It has long been known that the bathymodioline mussel gill has three types of proliferation cells that are conserved throughout all filibranchia bivalves: the budding zone at the posterior end of the gill where new gill filaments are continuously formed, and the dorsal and ventral ends of each gill filaments^{57,58}. These "cambial-zone"-like cell populations could continuously proliferate throughout the whole life span of the mussel⁵⁸. Our snRNA-seq data recognised three types of proliferation cells, which is consistent with previous findings (Fig. 4A, Fig. 4B and Supplementary Table S4). The gill posterior end budding zone cells (PEBZCs) are located on the first few freshly proliferated filaments of the posterior tip of gill (Figs. 4C and 4D). The PBEZCs marker (*Bpl_scaf_61993-0.4*) gradually disappeared at around the 11th -12th row of the gill filament (Fig. 4D), suggesting the maturation of the gill filaments, which is similar to the developmental pattern reported in another deep-sea mussel *Bathymodiolus azoricus*⁵⁹. The dorsal end proliferation cells (DEPCs), which expressed the hallmarks genes of muscular tissue (Fig. 4B), as well as cell proliferation and differentiation regulators, were the proliferation cells in connective tissue at the dorsal end of the gill slice (Fig. 4A, 4E). The ventral end proliferation cells (VEPCs) were two symmetrical triangle-shaped cell clusters of small symbiont-free cells (Fig. 4F). In VEPCs, genes encoding ribosomal proteins, chromatin proteins, RNA and DNA binding proteins, and cell proliferation markers were all up-regulated (Fig. 4B and Supplementary Table S4), indicating that VEPCs are meristem-like cells^{59,60}.

It has been hypothesised that new proliferation cells will be colonised by symbionts, serving as the vital mechanism of bacteriocyte recruitment⁵⁹. We then determined which proliferation cell type gives rise to the mature cells, especially the bacteriocytes, which has little available

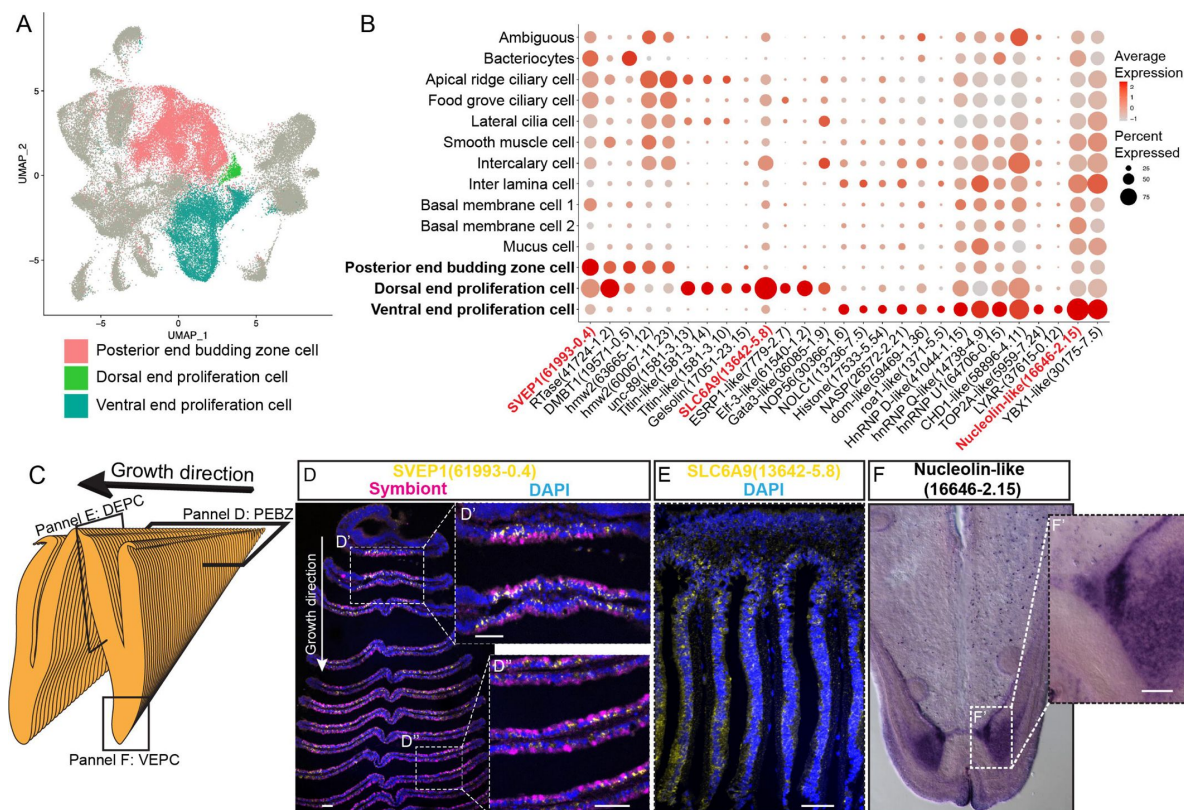


Figure 4

Proliferation cell populations of *G. platifrons* gill. A: UMAP representation of the three proliferation cell populations. B: Expression profiles of the cell markers that are specific or enriched in the supportive cell populations. The sizes of the circles represent the percentages of cells in those clusters that expressed a specific gene. Genes shown in red were validated by WISH or double FISH. C and D: Photographic and schematic analyses of the spatial position of the three Proliferation cell populations. E, F and J: FISH and WISH characterisation of the selected population markers. The marker genes confirmed by ISH or WISH in the current study are indicated in red. Scale bar: 50 μ m.

information regarding their precursor and transmission mode in bathymodioline mussels
[59](#), [61](#)

Slingshot trajectory analysis suggested that the PEBZCs might be the origin of all gill epithelial cells, including the other two proliferation cells (VEPC and DEPC) and bacteriocytes (Supplementary Fig. S6). The sole exception was BMC2, which may be derived from VEPC rather than PEBZC. This result is consistent with previous studies which suggested new gill filaments of the filibranch mussels are formed in the gill's posterior budding zones^{[59](#)}. The colonisation by the symbiont might play a crucial role in determining the fate of the bacteriocytes. Noticeably, in *G. platifrons*, only the pillar-shaped first row of gill filament, comprised of small meristem-like cells, was symbiont-free (**Figs. 4D and D**'), whereas all the other gill filaments were colonised by symbionts. This pattern of symbiosis establishment is different from that of *B. azoricus*, in which PEBZCs are symbiont free and are gradually colonised by symbionts released from the bacteriocytes on the adjacent mature gill filaments after maturation^{[59](#)}. On mature gill filaments, the DEPCs and VEPCs are seemingly the sources of new cells that sustain the growth of the gill filament from both dorsal and ventral directions, respectively^{[57](#)}. Interestingly, comparable active ventral and dorsal end proliferation zones have also been identified in the symbiotic mussel *B. azoricus*, whereas they are absent in the shallow-water mussel *Mytilus edulis*^{[62](#)}. This contrast further suggests the potential involvement of DEPCs and VEPCs in the establishment of symbiosis.

3. Bacteriocytes and host-symbiont interaction

We conducted the whole-mount FISH using a bacterial 16S rRNA probe for symbiont to determine the spatial distribution of bacteriocytes, and their positions relative to the other cell types on the gill filament (**Figs. 5A, B**). Bacteriocytes covered the majority of the surface of the gill filament, except the ventral tip, the ciliary disk, and the frontal edge (lateral ciliary). The bacteriocytes were surrounded by intercalary cells with microvilli and cilia on the surface (**Fig. 5C**).

The gene expression profile of bacteriocytes aligned well with the morphological analysis, which suggested that the bacteriocytes have structural, metabolic, and regulatory adaptations to cope with the symbiont (**Figs. 5D**, **5F** and 5I). It has been hypothesised that the bacteriocytes extract nutrients from the symbiont through two pathways: lysing the symbiont through the lysosome (the 'farming' pathway) or directly utilising the nutrient produced by the symbiont (the 'milking' pathway)^{[63](#), [64](#)}. Our TEM observations clearly detected intracellular vacuole and lysosome system of the bacteriocyte that harbour, transport, and digest the symbionts (**Fig. 5D**), which is consistent with previous studies^{[18](#), [65](#)}. The snRNA data showed bacteriocytes expressed cellular membrane synthesis enzyme (phosphoethanolamine N-methyltransferase, Bpl_scaf_15282-3.10) and a series of lysosomal proteins such as lysosomal proteases (cathepsins; Bpl_scaf_61711-5.12, Bpl_scaf_46838-6.40 and Bpl_scaf_59648-4.5; lysosomal alpha-glucosidase-like), protease regulators 56 (TMPRSS15 proteins; Bpl_scaf_52188-1.15 and Bpl_scaf_15410-0.9), and lysosomal traffic regulator proteins (rabenosyn-5, Bpl_scaf_54816-0.3 and Bpl_scaf_52809-1.6, lysosomal-trafficking regulator-like isoform X1). Among the proteases, cathepsins were thought to be evolutionary conserved molecular tools that host utilized to control the residence of their symbiont microbes^{[66](#)}. They were also highly expressed in symbiotic tissue of other deep-sea chemosynthetic animals, such as vesicomyid clams and vestimentiferan tubeworms^{[67](#)–[69](#)}. Bacteriocytes also expressed genes encoding cellular vesicle transports (kinesins, Bpl_scaf_14819-0.11, Bpl_scaf_54265-1.13, and Bpl_scaf_4784-1.40)^{[53](#)}, potential amino acid transporter^{[57](#)} (Bpl_scaf_36159-5.5), and genes involved in intracellular vesicle transport, such as the FYVE and coiled-coil domain-containing protein 1^{[58](#)} (Bpl_scaf_33726-5.7). In addition to form and mobilise early endosomes, the protein products of these genes could transport symbiont-secreted nutrient vesicles to the host (**Fig. 5F** and Supplementary Table S5), supporting the "milking" pathway.

The symbiont of *G. platifrons* belongs to type I methanotrophy, of which the core metabolic function is linked with the development of intracytoplasmic membranes leading to a high lipid/biomass content^{[70](#), [71](#)}. Recent lipid biomarker analyses showed that the gill of *G.*

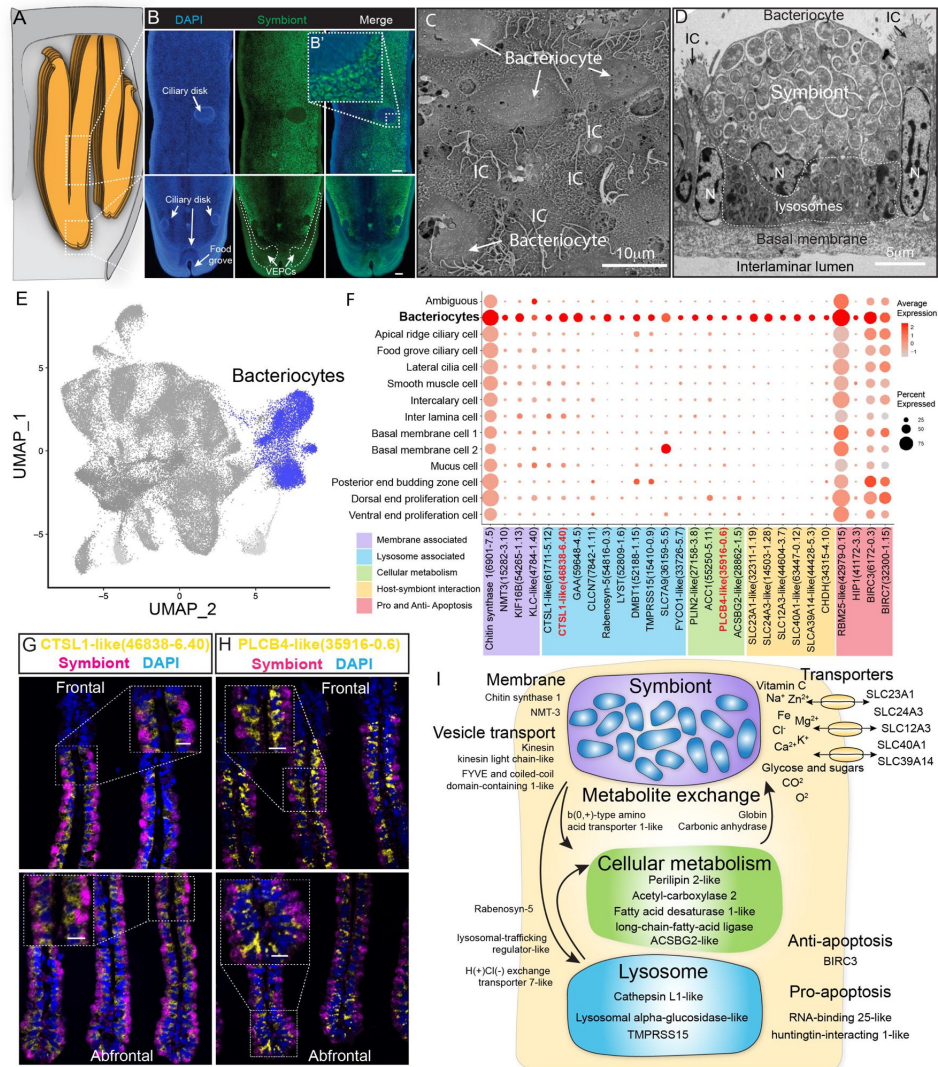


Figure 5

Characterisation of the bacteriocytes of *G. platifrons*. A: Schematic of the overall structure of *G. platifrons* gill filaments. B: Whole-mount FISH analyses of the overall distribution of bacteriocytes on the *G. platifrons* gill filament. C: SEM analysis of the bacteriocytes. D: TEM analysis of a bacteriocyte. E: UMAP representation of *G. platifrons* bacteriocytes. F: Expression profiles of the cell markers that are specific or enriched in the bacteriocytes. The sizes of the circles represent the percentages of cells in those clusters that expressed a specific gene. Double FISH validated the genes shown in red. I: Schematic of the host-symbiont interaction based on the single-cell transcriptome of *G. platifrons* bacteriocytes. The marker genes confirmed by ISH in the current study are indicated in red. Scale bar in panel G and H: 25 μ m.

platifrons contains a high amount of bacterial lipids, which are directly utilised by the host to synthesise most of its lipid contents⁶⁷. Downstream of the bacteriocyte's metabolic cascade, genes encoding proteins that may be involved in fatty acid/lipid metabolism, such as perilipin2 (Bpl_scaf_27158-3.8), which is the critical protein to form intracellular lipid droplets⁷², and a variety of fatty acid metabolism enzymes (acetyl coenzyme A, Bpl_scaf_55250-5.11⁷³; fatty acid desaturase 1-like isoform X1, Bpl_scaf_35916-0.6^{74,75}; long-chain fatty acid-ligase ACSBG2-like isoform X1, Bpl_scaf_28862-1.5^{76,77}), were up-regulated, suggesting that the fatty acid could be a major form of nutrients passing from the symbiont to the host mussel.

Additionally, bacteriocytes expressed several solute carriers, including sodium/ascorbate cotransporter (**Fig. 5I**, solute carrier family 23 members 1-like, Bpl_scaf_32311-1.19), sodium/potassium/calcium exchanger (sodium potassium calcium exchanger 3-like, Bpl_scaf_14503-1.28), sodium/chloride ion cotransporter (solute carrier family 12 member 3-like isoform X1, Bpl_scaf_44604-3.7), ferrous iron transporter (solute carrier family 40 member 1-like, Bpl_scaf_63447-0.12) and zinc transporter (zinc transporter ZIP14-like, Bpl_scaf_44428-5.3). The solute carriers are a large family of ATP-dependent transporters that shuttle a variety of small molecules across the cellular membrane⁷⁸. In several model symbiotic systems, solute carriers play a vital role in host-symbiont interaction by either providing the symbiont with substrates^{79,80} or transporting symbiont-produced nutrients to the host^{81–83}. Previous studies demonstrated that solute carrier genes were expanded in deep-sea chemosynthetic animals' genome⁸⁴ and highly expressed in symbiotic organs⁸⁵ including bathymodioline mussel's gill¹². In bathymodioline mussels, previous bulk RNA-seq studies detected up-regulated expression of a large variety of solute carriers in the gill, which is consistent with the present study, suggesting that solute carriers may play crucial roles in shuttling nutrients in and out of bacteriocytes and in maintaining the suitable intracellular micro-environment (such as the SLC23A1 and SLC39A14) for the symbiont^{86,87}.

4. Cell-type-specific response to environmental stresses

To examine cell-type-specific acclimatisation to environmental changes the expression profiles of differentially expressed genes (DEGs) were compared between the three states (Fanmao, starved and reconstituted animals) of samples collected in our *in situ* transplantation experiment (full lists of DEGs are shown in Supplementary Table S6-S18). Notably, for each state three mussels were processed but pooled for nuclei extraction before snRNA sequencing (see Method for detail). We tested our hypothesis here that at the starvation site, a relatively low concentration of methane shall upset symbiont metabolism and thus substantially affect symbiont-hosting bacteriocytes by assessing the transcriptional changes per cell type. We calculated the distances between the three sets of samples on the UMAP plot (**Fig. 6A**) estimating the centroid coordinates per sample per cell type, and calculating the Euclid distance between the centroid coordinates of each pair of states per cell type (Supplementary Table S19). The impact of starvation was variable across cell types, as reflected by the cross-state distances (**Fig. 6B**, green bar). Starvation resulted in the most significant transcriptional changes in bacteriocytes reflected by a large Fanmao-vs-starvation distance (2.3), followed by VEPC and inter lamina cells (2.1 and 1.4, respectively). On the other hand, starvation had little impact on the transcriptions of ciliary cells and most supportive cells such as the food groove ciliary cell, BMC2, and mucus cells (distances <0.5). **Fig 6B** also shows that after reconstitution (moving the mussels back to the methane-rich site Fanmao for 3 days), the expressional profile of bacteriocytes rapidly changed back, reflected by a large starvation-vs-reconstitution distance and much smaller Fanmao-vs-reconstitution distance (2.3 vs. 0.8; Suppl. Tab. S19). This result coincided with and was supported by our pseudo-time analysis for bacteriocytes (**Fig. 6C**), showing that bacteriocytes in the reconstitution are in intermediate and transitional states between Fanmao and starvation.

For the bacteriocytes population, we conducted cell trajectory pseudo-timing and detailed DEG analysis to interpret the mechanism of host-symbiont interactions. It is important to acknowledge that our sampling strategy might have limitations, as more closely spaced time points could

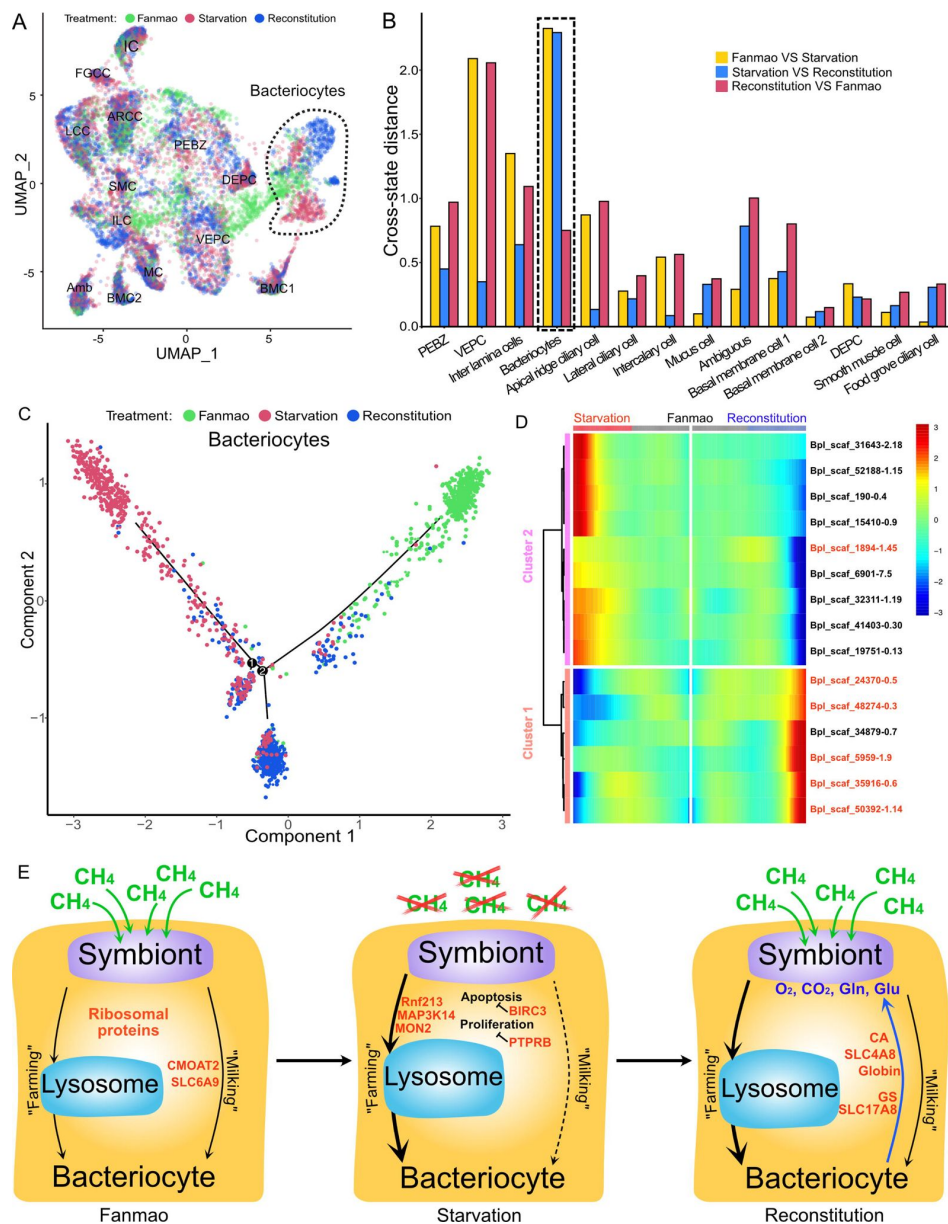


Figure 6

Analysis of cell population-specific DEGs. A: UMAP representation of the impact of deep-sea *in situ* transplant treatments on the gene expression pattern of each cell population. The cells from different treatments were labelled with different colours. The dashed line encircled bacteriocyte populations have a considerably altered expression profile. B: Histogram of cross-state distances between the centroids of the Fanmao, Starvation and Reconstitution groups per cell type on UMAP. The black dashed lines indicate the bacteriocyte populations whose expression profile was remarkably altered. C: Visualization of bacteriocytes onto the pseudo time map using Monocle. The black lines indicate the main path of the pseudotime ordering of the cells. D: Bifurcation of selected gene expression along two branches in response to environmental perturbation. Genes are clustered hierarchically into two groups, illustrating up- (cluster 1) and down- (cluster 2) regulated genes in the starvation state compared with Fanmao. Genes in red colour were discussed in [Section 4](#). The heat map showing the gene expression profiles of all bacteriocytes' DEGs is shown in supplementary Fig. S7; E: Proposed model for the molecular mechanisms of host-symbiont interactions in response to environmental changes.

enhance the confidence of trajectory reconstruction. The branched heatmap showed both up- and down-regulated genes during starvation and reconstitution compared with the Fanmao state (**Fig. 6D** and [supplementary Fig. S7](#)). The Kyoto Encyclopedia of Genes and Genomes (KEGG) pathway enrichment analysis of the bacteriocytes' DEGs provides an overall view of the pathways enriched in each environmental state (Supplementary Fig. S8 A-C). The genes encoding ribosomal proteins are highly expressed in the bacteriocytes under the methane-rich 'Fanmao' state (Supplementary Fig. S8A), suggesting an active protein synthesis and cellular metabolism ⁸⁸[88](#). Moreover, organic ion transporters (A BCB P-glycol, Bpl_scaf_18613-16.10; canalicular multispecific organic anion transporter 2, Bpl_scaf_52110-4.43; sodium- and chloride-dependent glycine transporter 1-like, Bpl_scaf_13642-5.8), which may be involved in transporting symbiont-produced nutrients, are also highly expressed (Supplementary Fig. S9).

In starved *G. platifrons*, negative regulators of cell proliferation, such as autocrine proliferation repressors and receptor-type tyrosine phosphatase beta, were up-regulated in bacteriocytes, suggesting repression of cell growth. We also observed the enrichment of genes in the apoptosis pathway (Supplementary Fig. S8B), which is attributed to the upregulation of caspases (Bpl_scaf_25165-3.8 and Bpl_scaf_24225-0.12) and cathepsins (Bpl_scaf_64706-0.16, Bpl_scaf_61711-5.12) which can trigger caspases-dependent cell death, suggesting cellular stress condition. Correspondingly, a gene encoding baculoviral IAP repeat-containing protein (Bpl_scaf_6172-0.3), which can bind caspase and inhibit apoptosis, was up-regulated in bacteriocytes (**Fig. 6E** [6E](#)). In *Drosophila*, the baculoviral IAP proteins are important in animal's response to cellular stress and promote cell survival^{89,90}[89](#)[90](#). Similarly, the gene encoding MAP3K14 (Bpl_scaf_60908-4.14) and potential E3 ubiquitin ligase RNF213 (Bpl_scaf_25983-1.26, Bpl_scaf_1894-1.45, Supplementary Fig. S10), which were apoptosis suppressor or regulator, were up-regulated⁹¹[91](#). These results suggest that starved bacteriocytes were carrying out cell-type-specific adjustments to cope with stresses. The E3 ubiquitin ligases could also work as intracellular immune sensors of bacterial lipopolysaccharides⁹²[92](#). Thus, the encoded protein may be able to active downstream immunological toolkit to digest the symbiont population for nutrients.

As mentioned above, bacteriocytes obtain nutrients from endosymbionts. KEGG pathway analyses suggested that phagosome, lysosome-related pathways were upregulated in the starvation state, indicating that bacteriocytes were actively digesting the endosymbionts in the starved *G. platifrons*. Interestingly, protein synthesis activity was more activated in inter lamina cells and VEPs as supported by the up-regulation of genes encoding ribosomal proteins in both cell populations (Supplementary Tables S6 and S17). This was contrary to the situation in bacteriocytes and may be a consequence of the high activity of "farming" in bacteriocytes.

We anticipated that after moving the starved *G. platifrons* back to Fanmao site, bacteriocytes and their endosymbionts would "reconstitute" leading to the partly restored "farming" and "milking" pathways. This was confirmed by higher expression of fatty acid metabolic genes in the mussels in the reconstitution state than those in the starvation state, such as long-chain-fatty-acid-ligase ACSBG2-like (Bpl_scaf_28862-1.5), elongation of very long-chain fatty acids 7-like (Bpl_scaf_5959-1.9), and fatty acid desaturase 1-like (Bpl_scaf_35916-0.6). These findings were also consistent with the result of KEGG analyses. The mitochondrial trifunctional enzyme (Bpl_scaf_42376-0.5) was also highly expressed, suggesting a high level of energy-producing activity. The KEGG analyses showed that glutamatergic synthase was enriched in the reconstitution state in comparison with those in both the Fanmao and starvation states. In the insect aphid-Buchnera endosymbiosis model, the host-produced glutamate could be transported and directly utilised by the symbiont⁹³[93](#). Similar glutamate-based host-symbiont metabolic interaction mechanisms were also proposed in the deep-sea mussel *Bathymodiolus thermophilus* and *B. azoricus*^{64,94}[64](#)[94](#). As the pseudotime analyses showed above (**Fig. 6C** [6C](#)), reconstitution was the intermediate state between Fanmao and starvation. Metabolic and gene regulatory functions of bacteriocytes were still different from that in Fanmao. The KEGG analyses suggested regulatory (Rap1 signalling, retrograde endocannabinoid signalling, chemokine signalling, glucagon signalling, thyroid hormone synthesis, mRNA

surveillance pathway, etc.) and metabolic pathways were enriched in the reconstitution state (fatty acid metabolism, salivary secretion, aldosterone synthesis and secretion, insulin secretion and pancreatic secretion) (Supplementary Fig. S8C). For example, we detected up-regulation of the genes encoding carbonic anhydrases (Bpl_scaf_33596-7.3 and Bpl_scaf_48274-0.3), electroneutral sodium bicarbonate exchanger 1 (Bpl_scaf_61230-5.11), and globin-like proteins (Bpl_scaf_50392-1.14 and Bpl_scaf_24370-0.5), which could provide the symbiont with carbon dioxide and oxygen necessary for the symbiont's chemosynthetic metabolism^{84,95}.

5. Summary and outlook

Using the deep-sea mussel *G. platifrons* as a model organism, we demonstrated the power of integrating snRNA-seq and WISH data in unravelling the mechanism behind animal-microbe symbiosis. The robustness of this strategy showed stable and highly distinguishable expression patterns of each cell type regardless of the different environmental states. We successfully profiled the specific roles of different types of cells, including the previously unknown cell types, in maintaining the structure and function of the gill. The supportive cells that located in between (inter lamina cells) and on the basal membrane's surface (BMC1 and BMC2) helped maintain the anatomical structure of the basal membrane. Ciliary and smooth muscle cells involved in gill slice contraction and cilium beat allowing the gathering of material and information from the surrounding environment. Proliferation cells (PBEZCs, DEPCs and VEPCs) gave rise to new cells, including bacteriocytes which obtained nutrients from endosymbionts using intracellular vacuoles and lysosomes.

The snRNA analyses also revealed that different cell types collaborated to support bacteriocytes' functionality. The PBEZCs gave rise to new bacteriocytes that allowed symbiont colonisation. Bacteriocytes attached to the basal membrane and were stabilised by supportive cells. Mucus cells co-localise with bacteriocytes and help to maintain immune homeostasis. The beating ciliary cells controlled the water flow, providing bacteriocytes with necessary inorganic substances from the environment. This new information on cell-cell interaction certainly advanced our overall understanding of how endosymbiotic microbes and host cells communicate and collaborate, which cannot be easily achieved through other methods.

Moreover, the analysis of snRNA data from bacteriocytes has provided insight into the molecular mechanisms employed by the host to maintain and regulate the symbiosis. Notably, the bacteriocytes-enriched transcripts involved in harbouring, digesting symbionts, and transporting nutrients produced by the symbionts were identified and characterised. Our *in situ* transplant experiments by moving mussels between methane-rich and methane-limited sites also provided clues of cell-type specific responses to environmental change. Under a methane-limited environment, the starved mussels more actively consumed endosymbionts through the "farming" pathway. After being moved back to the methane-rich environment, mussels produced more glutamates which sustained the regrowth of symbionts. These preliminary findings showed that the deep-sea mussels were able to control their endosymbionts using a set of genes in response to environmental change. Due to the limitation of remotely operated vehicle (ROV) dives and sampling capacity, we had to have pooled samples of each state for nuclei extraction and sequencing. Thus, the cells per cluster could be considered as technical than biological replicates. Although this sampling process strategy has been broadly used in snRNA/scRNA sequencing⁹⁶, we recognise the possible violation of assumptions in *p*-value calculation for DEGs between the three states.

Overall, the single-cell spatial and functional atlas developed in the present work will decipher some common principles of symbiosis and environmental adaptation mechanisms of animals. The workflow developed in the present study could provide insightful references for researchers focusing on the mechanistic study of the biological adaptation of biologically and ecologically important non-model animals.

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Author contributions

W.H., K.C., P.Y.Q. and L.C.L. designed the study. Z.H., L. Cao, Z.Z.S., L. Chao and W.M.X. conducted the *in situ* transplant experiments and collected the samples; Z.Q.Y. and W.H. constructed the snRNA-seq library. H.K. performed the bioinformatics analysis. W.H., Z.H., L. Cao, J.L., H.C. and L.Z. performed the experiments. This paper was written by W.H., H.K., K.C., P.Y.Q. and L.C.L. All authors revised the manuscript. All authors read, approved, and contributed to the final manuscript.

Competing interests

The authors declare no competing interests.

Materials and methods

Deep-sea *in situ* transplant experiment and sample collection

In situ transplant experiment was conducted at the 'F-site' cold seep during the *R/V 'Kexue'* 2020 South China Sea cold seep cruise. The overall design of the *in situ* transplant experiment and environmental states are shown in **Figure 1** [↗](#). First, mussels in the methane-rich 'Fanmao' site (meaning 'prosperous' site, 22°06.55.425'N, 119°17.08.287'E, depth 1117m) were scoped into three nylon bags with approximately 10 mussels in each bag. Then, two bags of mussels were transplanted to the low methane Starvation site (**Figure 1** [↗](#), 22°07.00'N, 119°17.07.02'E, depth 1147.42m). After 11 days of transplantation, one bag of mussels in the Starvation site was moved back to the 'Fanmao' site. On the 14th day of the transplant, three bags of mussels: one bag from the Fanmao site (designated as the 'Fanmao' sample), one bag from the starvation site (designated as the 'Starvation' sample) and one bag of mussels which were first transplanted to the Starvation site for 11 days and moved back to the Fanmao site for 3 days (designated as the 'Reconstitution' sample) were all retrieved by ROV Faxian. The mussels were kept in a hydraulic pressure-sealed biobox during the ascending of the ROV. The biobox is made of heat-insulated material, which will prohibit heat exchange with warm surface water. The samples were immediately processed once onboard *R/V Kexue*. For snRNA-seq, the posterior end tip of the mussel's gill was dissected (**Figure 1C** [↗](#)), snap-frozen with liquid nitrogen and then stored at -80 °C until use. For WISH, ISH and FISH, the gills of the mussels were first fixed with 4% paraformaldehyde (PFA, prepared with autoclaved 0.22 μM membrane filtered *in situ* seawater) at 4 °C overnight. Then, the gill was washed with ice-cold 1× PBS three times, dehydrated and stored in 100% methanol at -20 °C.

Single-nucleus RNA-sequencing of *G. platifrons*

The gill nucleus was extracted using the Nuclei PURE prep nuclei isolation kit (Sigma-Aldrich). For each sample, posterior end tips from 3 individual mussels were randomly selected and pooled together. The posterior tips were homogenised in 10 mL of ice-cold lysis solution (Nuclei PURE lysis buffer with 0.1% Triton X100 and 1 mM DTT) on ice. The cell nuclei were then separated by sucrose gradient centrifugation according to the manufacturer's protocol. The cell nuclei pellets were washed by re-suspending in icecold DPBS–BSA solution (1× DPBS, 0.04% nuclease-free BSA, 0.01% RNase inhibitor; Takara). The nucleus was spun down by centrifugation at 500×g for 5 min at 4 °C. This step was repeated to remove the contaminants from the cell plasma. Finally, the nucleus was re-suspended in the DPBS–BSA solution. The concentration of the cell nucleus was counted by Cell Countess II. Single-nucleus RNA-seq libraries were then constructed with the BD Rhapsody™ single-cell analysis system using the BD Rhapsody WTA Amplification Kit according to the manufacturer's protocol. The library was subjected to 150bp paired-end sequencing using the Illumina HiSeq 4000 plat-form. The clean reads of three datasets were submitted to the National Centre for Biotechnology Information Sequence Read Archive database (Bioproject: PRJNA779258).

Bioinformatics

The three raw datasets (Fanmao, Starvation, and Reconstitution) were processed individually following the BD single-cell genomics analysis setup user guide (Doc ID: 47383 Rev. 8.0) and BD single-cell genomics bioinformatics handbook (Doc ID: 54169 Rev. 7.0). This process involved the preparation of gene names, alignment of data to the genome, and generation of an expression matrix for each dataset. A *G. platifrons* genome reference v1.0 (available at the Dryad Digital Repository; <http://dx.doi.org/10.5061/dryad.h9942>) was utilized. The genome was indexed and constructed using STAR 2.5.2b⁹⁷. Subsequently, the sequencing data was mapped to the indexed genome using the BD Rhapsody single-nucleus pipeline v1.9, employing default parameters.

Each data matrix was converted to a SingleCellExperiment data format, and empty barcodes were removed using the emptyDrops function of DropletUtils v3.14⁹⁸. Then, we converted and processed the data in Seurat v3²⁸. We removed cells that had <100 or > 2500 genes and <100 or > 6000 unique molecular identifiers (UMIs). We also removed genes that had <10 UMIs in each data matrix. Then, we log-normalised the data and used DoubletFinder v2.0⁹⁹ to remove potential doublet, assuming a 7.5% doublet formation rate. The numbers of retained nuclei were 9717, 21614 and 28928 for Fanmao, Starvation and Reconstitution data, respectively (Supplementary Table S20). We used the top 3000 highly variable genes for principal component analysis (PCA) and a reciprocal PCA approach to integrating the three datasets¹⁰⁰. The first 40 principal components (PCs) were used for uniform manifold approximation and projection (UMAP) dimensional reduction and following clustering (Supplementary Fig. S11). We employed an empirical parameter - a resolution of 0.2 - in the FindClusters function, utilizing the original Louvain algorithm. To identify unique marker genes associated with each cluster, we utilized the FindAllMarkers function from the Seurat package. This analysis employed the Wil-coxon rank sum test, focusing on genes that demonstrated a minimum 0.3-fold difference between two groups of cells. The annotation of cell types relied on the reference of previously published marker genes. In instances where clusters exhibited marker genes that couldn't be associated with known cell types, we pursued the validation process through whole-mount *in situ* hybridization (WISH; elaborated below), discerning cell types based on gene expression patterns and morphological characteristics. When WISH displayed uniform expression of marker genes from different clusters within the same cell type, we consolidated those clusters into a single cell type. We assigned cells to 14 reliable cell types (Supplementary Fig. S11). Supplementary data S1 presents the counting matrix and Supplementary data S1 presents the average expression of each gene per cell type. To evaluate the stability of each cell type, we implemented a bootstrap sampling and clustering strategy comprising 100 iterations using cells combined from all three samples and individual

samples¹⁰¹. The determination of marker genes per cell type followed the previously described methodology. The identified marker genes were subsequently utilized in KEGG enrichment analysis, employing ClusterProfiler v4.2¹⁰².

Cell trajectory analysis

We conducted slingshot trajectory analyses using Slingshot¹⁰³ v2.2.1 for a subset of clusters (intercalary cells, dorsal-end proliferation cells, ventral-end proliferation cells, mucus cells, basal membrane cells 1, basal membrane cells 2 and bacteriocytes) to explore the developmental trajectory of cells, assuming that all these cells were developed from the same precursor ('posterior-end budding zone'). We performed PCA for the subsampled data set and conducted dimensional reduction using phateR v1.0.7 for slingshot analyses. We used the Potential of Heat-diffusion for Affinity-based Trajectory Embedding (PHATE) because it could better reveal developmental branches than other tools¹⁰⁴.

We also examined the effect of deep-sea transplant experiments on shaping gene expression patterns by comparing the expression levels amongst the three different states of a given cluster. We conducted Monocle analyses using Monocle2 and Monocle 3 in R environment^{105,106}. This comparison was done for the bacteriocytes. The biased number of cells per state could affect the results of the dimensional reduction and calculation of marker genes, and the sequenced nucleus per state was unbalanced; therefore, we first downsampled the cells per cluster per state to a maximum of 1000 nuclei per cell type. Thereafter, we performed PCA and dimensional reduction using UMAP and PHATE, calculated marker genes and conducted slingshot trajectory and KEGG enrichment analyses as mentioned above for each cell type.

Phylogenetic estimation

Phylogenetic estimation was conducted for E3 ubiquitin ligase RNF213 genes. We downloaded RNF213 genes from GenBank for representative vertebrate and invertebrate species across the tree of life. The amino acid sequences were aligned with the *G. platifrons* sequences annotated as E3 ubiquitin ligase (Bpl_scaf_25983-1.26 and Bpl_scaf_1894-1.45) using MAFFT v7.450. The alignment was used to estimate a maximum-likelihood best gene tree and to calculate bootstrap values on each node using RAxML v8.2.

G. platifrons gill fixation and storage

The gill tissues of *G. platifrons* were dissected within minutes after the ROV, and samples were retrieved on board R/V Kexue. The gill tissues were briefly washed with ice-cold filtered and autoclaved *in situ* seawater (FAISW) and then fixed in 4%

PFA prepared in FAISW at 4 °C overnight. The gill tissues were washed three times with ice-cold PBST, dehydrated in 100% methanol and stored at -20 °C until use.

Synthesise probes for mRNA *in situ* hybridisations

For WISH and double FISH analyses, the DNA fragments (~1000bp) of the targeted genes were first PCR amplified with gene-specific primers (GSPs) pairs using *G. platifrons* gill cDNA as template (the sequences of targeted genes and gene-specific primers were provided in Supplementary Data S3). The amplified fragments were ligated into the pMD18-simpleT vector (Takara) and transformed into *E. coli*. Individual colonies were picked up, and their plasmids were sequenced to confirm the inserts. The templates for *in vitro* mRNA transcription were amplified using T7 forward GSP (sense probe control) or Sp6 reversed GSPs (antisense probe) combined with either forward or reversed genespecific primer. Labelled probes and control probes were generated using digoxigenin (DIG)-12-UTP (Roche) or fluorescein-12-UTP (Roche) according to the protocol described by Thisse¹⁰⁷ with Sp6 and T7 RNA polymerase, respectively.

Paraffin embedding and double fluorescent *in situ* hybridisation

The methanol-dehydrated gill slices were incubated in 100% ethanol, a 1:1 mixture of 100% ethanol and xylene and Xylene twice for 1 h each at RT. The samples were embedded by incubating in Paraplast Plus (Sigma-Aldrich) for 2 h at 65 °C and then cooled down to RT. Sections with 5 µm thickness were cut using a microtome (Leica).

For double FISH, sections were dewaxed by incubating in xylene twice, a 1:1 mixture of 100% ethanol and xylene, 100% ethanol twice, 95% ethanol, 85% ethanol and 75% ethanol for 15 min each at RT. The sections were washed with PBST three times for 10 min each and then permeabilised by 2 µg/mL proteinase K (NEB) in PBST for 15 min at RT. Post-digestion fixation was conducted by incubating the sections in 4% PFA in PBST for 30 min at RT. The sections were washed three times with PBST for 15 min each. Pre-hybridisation was conducted by incubating the sections in HM for 1 h at 55 °C. Then, the *in situ* hybridisation was performed by incubating the sections in ~0.5 ng/µL fluorescein (Roche)-labelled probe prepared in fresh HM overnight at 55 °C. The sections were washed three times with 2×SSC for 15 min each at 55 °C, cooled down to room temperature and washed three times with PBST.

A second-round hybridisation was conducted on the DIG-labelled oligonucleotides to label the symbiont. The slices were hybridised for 1 h at 46 °C with 100 ng DIG-labelled *G. platifrons* symbiont-specific probe in FISH buffer (0.9 M NaCl, 0.02 M Tris-HCl, 0.01% sodium dodecyl sulphate [SDS] and 30% formamide). The slices were then washed with FISH washing buffer (0.1 M NaCl, 0.02 M Tris HCl, 0.01% SDS and 5 mM EDTA) three times at 5 min each at 48 °C. The slices were washed with PBST three times and then blocked with blocking buffer (2.5% sheep serum and 2% BSA in sheep serum) for 1 h at room temperature.

The slices were then incubated with 1:1000 diluted anti-fluorescein–peroxidase (POD) (Roche) overnight at 4 °C, then washed six times with PBST for 15 min each and three times with TNT buffer (100 mM Tris-HCl, pH 7.5; 100 mM NaCl; 0.1% Tween 20) for 15 min each. Afterwards, the fluorescent signal of the *G. platifrons* gene expression pattern was developed by the TSA fluorescein kit (Akoya Biosciences) according to the manufacturer's protocol. The slices were washed three times, and the remaining POD activity was quenched by incubation in 1% hydrogen peroxide solution for 1 h at room temperature. Then, the slices were washed three times with PBS, blocked with blocking buffer for 30 min at room temperature, incubated with 1:2500 diluted anti-DIG–POD (Roche) for 2 h at RT and washed with PBST for six times and TNT for three times. The symbiont FISH signal was developed using the TSA Cy3 kit (Akoya Biosciences). Finally, the slices were washed with PBST, stained with DAPI, and mounted with ProLong Diamond Antifade Mountant (Thermo Fisher).

Whole-mount *in situ* hybridisation

For WISH, the connective region at the end of the W-shaped gill filament was cut off, and each gill slice was carefully peeled off with fine-tip tweezers. We dissected gill tissues from 5 individual mussels and pooled all the gill slices together. The gill slices were then rehydrated in 75%, 50% and 25% methanol-PBST (1×PBS with 0.1% Tween 20) for 15 min each, followed by 3×5 min PBST washes. The gill slices were then permeabilised with 2 µg/mL proteinase K in PBST for 30 min at 37 °C. Post-digestion fixation was conducted by fixing the gill slices with 4% PFA in PBST for 30 min at room temperature (RT). After 3×5 min PBST wash to remove the residual fixative, the gill slices were pre-hybridised with a hybridisation mix (HM; containing 50% formamide, 5×saline–sodium citrate (SSC), 0.1% Tween 20, 10 µg/mL heparin, 500 µg/mL yeast tRNA) for 1 h at 65 °C. For each hybridisation, 5–10 gill slices were added to 400 µL of fresh HM containing ~0.5 ng/µL DIG-labelled probe. Hybridisation was conducted in a 55 °C shaking water bath overnight. Post-hybridisation washes were performed according to the following steps: the gill slices were first washed 3×15 min with hybridisation washing buffer (50% formamide, 5×SSC, 0.1% Tween 20), followed by 3×15 min

2×SSC with 0.1% Tween 20 and 3×15 min 0.2×SSC with 0.1% Tween 20. The washings were also conducted in a shaking water bath, and all the washing buffers were pre-heated to 55 °C. The samples were washed three times with PBST and then blocked in blocking buffer (2.5% sheep serum, 2% BSA in PBST) for 1 h at RT. Each sample was incubated with 1:10,000 diluted anti-DIG-AP antibody (Roche) at 4 °C overnight. The samples were incubated with the antibody, then washed for 6×15 min with PBST, followed by washing in 3×15 min alkaline Tris buffer (100 mM Tris-HCl, pH 9.5; 100 mM NaCl; 50 mM MgCl₂). The samples were incubated in nitro blue tetrazolium/5-Bromo-4-chloro-3-indolyl phosphate staining solution (Sangon). After the desired expression pattern was revealed, the staining reaction was stopped by 3×15 min PBST-EDTA wash (PBST, 1 mM EDTA). The gill slices were cleared by incubating in 100% glycerol overnight at 4 °C and then mounted on glass slides. The results of control hybridisations (with sense probes) were provided in Supplementary Fig. S13-S16. WISH analyses were repeated with another batch of *G. platifrons* gill slices samples collected during the *R/V 'Kexue'* 2017 South China Sea cold seep cruise to confirm the consistency in expression patterns.

Microscopy imaging

All the WISH samples and whole-mount 16S FISH images were observed and imaged with a Nikon Eclipse Ni microscope with a DS-Ri2 camera. The double FISH slides were imaged with a Zeiss LSM710 confocal microscope.

Electron microscopy analysis

The gill slices of the *G. platifrons* were dissected and fixed in electron microscopy fixative (2.5% glutaraldehyde and 2% PFA) at 4 °C. For SEM analysis, the samples were dehydrated in a graded ethanol series and then dried at the critical point. The samples were then coated with gold (sputter/carbon Thread, EM ACE200) and observed under a scanning electron microscope (VEGA3, Tescan). For TEM analysis, the samples were rinsed with double distilled water, post-fixed with 1% osmium tetroxide and then washed with double distilled water. The samples were then rinsed, dehydrated and embedded in Ep812 resin. Ultra-thin sections were obtained with an ultramicrotome (70 nm thickness, Reichert-Jung Ultracut E). The sections were then double-stained with lead citrate and uranyl acetate. The cells were observed under a transmission electron microscope (JEM1200, Jeol) operated under 100 kV.

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Reviewer #1 (Public Review):

Wang, He et al have constructed comprehensive single nucleus atlas for the gills of the deep sea Bathymodioline mussels, which possess intracellular symbionts that provide a key source of carbon and allow them to live in these extreme environments. They provide annotations of the different cell states within the gills, shedding light on how multiple cell types cooperate to give rise to the emergent functions of the composite tissues and the gills as a whole. They pay special attention to characterizing the bacteriocyte cell populations and identifying sets of genes that may play a role in their interaction with the symbiotes.

Wang, He et al sample mussels from 3 different environments: animals from their native methane rich environment, animals transplanted to a methane-poor environment to induce starvation and animals that have been starved in the methane-poor environment and then moved back to the methane-rich environment. They demonstrated that starvation had the biggest impact on bacteriocyte transcriptomes. They hypothesize that the up-regulation of genes associated with lysosomal digestion leads to the digestion of the intracellular symbiont during starvation, while the non-starved and reacclimated groups more readily harvest the nutrients from symbiotes without destroying them. Further work exploring the differences in symbiote populations between ecological conditions will further elucidate the dynamic relationship between host and symbiote. This will help disentangle specific changes in transcriptomic state that are due to their changing interactions with the symbiotes from changes associated with other environmental factors.

This paper makes available a high quality dataset that is of interest to many disciplines of biology. The unique qualities of this non-model organism and collection of conditions sampled make it of special interest to those studying deep sea adaptation, the impact of environmental perturbation on Bathymodioline mussels populations, and intracellular symbiotes. The authors also use a diverse array of tools to explore and validate their data.

<https://doi.org/10.7554/eLife.88294.2.sa2>

Reviewer #2 (Public Review):

Wang, He et al. shed insight into the molecular mechanisms of deep-sea chemosymbiosis at the single-cell level. They do so by producing a comprehensive cell atlas of the gill of *Gigantidas platifrons*, a chemosymbiotic mussel that dominates the deep-sea ecosystem. They uncover novel cell types and find that the gene expression of bacteriocytes, the symbiont-hosting cells, supports two hypotheses of host-symbiont interactions: the "farming" pathway, where symbionts are directly digested, and the "milking" pathway, where nutrients released by the symbionts are used by the host. They perform an in situ transplantation experiment in the deep sea and reveal transitional changes in gene expression that support a model where starvation stress induces bacteriocytes to "farm" their symbionts, while recovery leads to the restoration of the "farming" and "milking" pathways.

A major strength of this study includes the successful application of advanced single nucleus techniques to a non-model, deep sea organism that remains challenging to sample. I also applaud the authors for performing an in situ transplantation experiment in a deep sea environment. From gene expression profiles, the authors deftly provide a rich functional description of *G. platifrons* cell types that is well-contextualized within the unique biology of chemosymbiosis. These findings offer significant insight into the molecular mechanisms of deep-sea host-symbiont ecology, and will serve as a valuable resource for future studies into the striking biology of *G. platifrons*.

The authors' conclusions are generally well-supported by their results. However, I recognize that the difficulty of obtaining deep-sea specimens may have impacted experimental design and no replicates were sampled.

It is notable that the Fanmao cells were much more sparsely sampled. It appears that fewer cells were sequenced, resulting in the Starvation and Reconstitution conditions having 2-3x more cells after doublet filtering. These discrepancies also are reflected in the proportion of cells that survived QC, suggesting a distinction in quality or approach. However, the authors provide clear and sufficient evidence via bootstrapping that batch effects between the three samples are negligible. While batch effect does not appear to have affected gene expression profiles, the proportion of cell types may remain sensitive to sampling techniques, and thus interpretation of Fig. S12 must be approached with caution.

<https://doi.org/10.7554/eLife.88294.2.sa1>

Reviewer #3 (Public Review):

Wang et al. explored the unique biology of the deep-sea mussel *Gigantidas platifrons* to understand fundamental principles of animal-symbiont relationships. They used single-nucleus RNA sequencing and validation and visualization of many of the important cellular and molecular players that allow these organisms to survive in the deep-sea. They demonstrate that a diversity of cell types that support the structure and function of the gill including bacteriocytes, specialized epithelial cells that host sulfur-oxidizing or methane-oxidizing symbionts as well as a suite of other cell types including supportive cells, ciliary, and smooth muscle cells. By performing experiments of transplanting mussels from one habitat which is rich in methane to methane-limited environments, the authors showed that starved mussels may consume endosymbionts versus in methane-rich environments upregulated genes involved in glutamate synthesis. These data add to the growing body of literature that organisms control their endosymbionts in response to environmental change.

The conclusions of the data are well supported. The authors adapted a technique that would have been technically impossible in their field environment by preserving the tissue and then performing nuclear isolation after the fact. The use of single-nucleus sequencing opens the possibility of new cellular and molecular biology that is not possible to study in the field. Additionally, the in-situ data (both WISH and FISH) are high-quality and easy to interpret. The use of cell-type-specific markers along with a symbiont-specific probe was effective. Finally, the SEM and TEM were used convincingly for specific purposes in the case of showing the cilia that may support water movement.

The one particular area for future exploration surrounds the concept of a proliferative progenitor population within the gills. The authors recover molecular markers for these putative populations and additional future work will uncover if these are indeed proliferative cells contribute to symbiont colonization.

Overall the significance of this work is identifying the relationship between symbionts and bacteriocytes and how these host bacteriocytes modulate their gene expression in response to environmental change. It will be interesting to see how similar or different these data are across animal phyla. For instance, the work of symbiosis in cnidarians may converge on similar principles of there may be independent ways in which organisms have been able to solve these problems.

<https://doi.org/10.7554/eLife.88294.2.sa0>

Author response:

The following is the authors' response to the original reviews.

eLife assessment

*This study provides an important cell atlas of the gill of the mussel *Gigantidas platifrons* using a single nucleus RNA-seq dataset, a resource for the community of scientists studying deep sea physiology and metabolism and intracellular host-symbiont relationships. The work, which offers solid insights into cellular responses to starvation stress and molecular mechanisms behind deep-sea chemosymbiosis, is of relevance to scientists interested in host-symbiont relationships across ecosystems.*

Public Reviews:

Reviewer #1 (Public Review):

Wang et al have constructed a comprehensive single nucleus atlas for the gills of the deep sea Bathymodioline mussels, which possess intracellular symbionts that provide a key source of carbon and allow them to live in these extreme environments. They provide annotations of the different cell states within the gills, shedding light on how multiple cell types cooperate to give rise to the emergent functions of the composite tissues and the gills as a whole. They pay special attention to characterizing the bacteriocyte cell populations and identifying sets of genes that may play a role in their interaction with the symbiotes.

Wang et al sample mussels from 3 different environments: animals from their native methane-rich environment, animals transplanted to a methane-poor environment to induce starvation, and animals that have been starved in the methane-poor environment and then moved back to the methane-rich environment. They demonstrated that starvation had the biggest impact on bacteriocyte transcriptomes. They hypothesize that the upregulation of genes associated with lysosomal digestion leads to the digestion of the intracellular symbiont during starvation, while the non-starved and reacclimated groups more readily harvest the nutrients from symbiotes without destroying them.

Strengths:

This paper makes available a high-quality dataset that is of interest to many disciplines of biology. The unique qualities of this non-model organism and the collection of conditions sampled make it of special interest to those studying deep sea adaptation, the impact of environmental perturbation on Bathymodioline mussels populations, and intracellular symbiotes. The authors do an excellent job of making all their data and analysis available, making this not only an important dataset but a readily accessible and understandable one.

The authors also use a diverse array of tools to explore their data. For example, the quality of the data is augmented by the use of in situ hybridizations to validate cluster identity and KEGG analysis provides key insights into how the transcriptomes of bacteriocytes change.

The authors also do a great job of providing diagrams and schematics to help orient non-mussel experts, thereby widening the audience of the paper.

Thank the reviewer for the valuable feedback on our study. We are grateful that the reviewers found our work to be interesting and we appreciate their thorough evaluation of our research. Their constructive comments will be considered as we continue to develop and improve our study.

Weaknesses:

One of the main weaknesses of this paper is the lack of coherence between the images and the text, with some parts of the figures never being referenced in the body of the text. This makes it difficult for the reader to interpret how they fit in with the author's discussion and assess confidence in their analysis and interpretation of data. This is especially apparent in the cluster annotation section of the paper.

We appreciate the feedback and suggestions provided by the reviewer, and we have revised our manuscript to make it more accessible to general audiences.

Another concern is the linking of the transcriptomic shifts associated with starvation with changes in interactions with the symbiotes. Without examining and comparing the symbiote population between the different samples, it cannot be concluded that the transcriptomic shifts correlate with a shift to the 'milking' pathway and not other environmental factors. Without comparing the symbiote abundance between samples, it is difficult to disentangle changes in cell state that are due to their changing interactions with the symbiotes from other environmental factors.

We are grateful for the valuable feedback and suggestions provided by the reviewer. Our keen interest lies in understanding symbiont responses, particularly at the single-cell level. However, it's worth noting that existing commercial single-cell RNA-seq technologies rely on oligo dT priming for reverse transcription and barcoding, thus omitting bacterial gene expression information from our dataset. We hope that advancements in technology will soon enable us to perform an integrated analysis encompassing both host and symbiont gene expression.

Additionally, conclusions in this area are further complicated by using only snRNA-seq to study intracellular processes. This is limiting since cytoplasmic mRNA is excluded and only nuclear reads are sequenced after the organisms have had several days to acclimate to their environment and major transcriptomic shifts have occurred.

We appreciate the comments shared by the reviewer and agree that scRNA-seq provides more comprehensive transcriptional information by targeting the entire mRNA of the cell. However, we would like to highlight that snRNA-seq has some unique advantages over scRNA-seq. Notably, snRNA-seq allows for simple snap-freezing of collected samples, facilitating easier storage, particularly for samples obtained during field trips involving deep-sea animals and other ecologically significant non-model animal samples. Additionally, unlike scRNA-seq, snRNA-seq eliminates the need for tissue dissociation, which often involves prolonged enzymatic treatment of deep-sea animal tissue/cells under atmospheric pressure. This process can potentially lead to the loss of sensitive cells or alterations in gene expression.

Moreover, snRNA-seq procedures disregard the size and shape of animal cells, rendering it a superior technology for constructing the cell atlas of animal tissues. Consequently, we assert that snRNA-seq offers flexibility and represents a suitable choice for the research objects of our current research.

Reviewer #2 (Public Review):

*Wang, He et al. shed insight into the molecular mechanisms of deep-sea chemosymbiosis at the single-cell level. They do so by producing a comprehensive cell atlas of the gill of *Gigantidas platifrons*, a chemosymbiotic mussel that dominates the deep-sea ecosystem. They uncover novel cell types and find that the gene expression of bacteriocytes, the symbiont-hosting cells, supports two hypotheses of host-symbiont interactions: the "farming" pathway, where symbionts are directly digested, and the "milking" pathway, where nutrients released by the symbionts are used by the host. They perform an in situ transplantation experiment in the deep sea and reveal transitional changes in gene expression that support a model where starvation stress induces bacteriocytes to "farm" their symbionts, while recovery leads to the restoration of the "farming" and "milking" pathways.*

*A major strength of this study includes the successful application of advanced single-nucleus techniques to a non-model, deep-sea organism that remains challenging to sample. I also applaud the authors for performing an in situ transplantation experiment in a deep-sea environment. From gene expression profiles, the authors deftly provide a rich functional description of *G. platifrons* cell types that is well-contextualized within the unique biology of chemosymbiosis. These findings offer significant insight into the molecular mechanisms of deep-sea host-symbiont ecology, and will serve as a valuable resource for future studies into the striking biology of *G. platifrons*.*

The authors' conclusions are generally well-supported by their results. However, I recognize that the difficulty of obtaining deep-sea specimens may have impacted experimental design. In this area, I would appreciate more in-depth discussion of these impacts when interpreting the data.

Thank the reviewer for their valuable feedback on our study. We're grateful that the reviewers found our work interesting, and we appreciate their thorough evaluation of our research. We'll consider their constructive comments as we continue to develop and improve our study.

Because cells from multiple individuals were combined before sequencing, the in situ transplantation experiment lacks clear biological replicates. This may potentially result in technical variation (ie. batch effects) confounding biological variation, directly impacting the interpretation of observed changes between the Fanmao, Reconstitution, and Starvation conditions. It is notable that Fanmao cells were much more sparsely sampled. It appears that fewer cells were sequenced, resulting in the Starvation and Reconstitution conditions having 2-3x more cells after doublet filtering. It is not clear whether this is due to a technical factor impacting sequencing or whether these numbers are the result of the unique biology of Fanmao cells. Furthermore, from Table S19 it appears that while 98% of Fanmao cells survived doublet filtering, only ~40% and ~70% survived for the Starvation and Reconstitution conditions respectively, suggesting some kind of distinction in quality or approach.

There is a pronounced divergence in the relative proportions of cells per cell type cluster in Fanmao compared to Reconstitution and Starvation (Fig. S11). This is potentially a very interesting finding, but it is difficult to know if these differences are the expected biological outcome of the experiment or the fact that Fanmao cells are much more

sparingly sampled. The study also finds notable differences in gene expression between Fanmao and the other two conditions- a key finding is that bacteriocytes had the largest Fanmao-vs-starvation distance (Fig. 6B). But it is also notable that for every cell type, one or both comparisons against Fanmao produced greater distances than comparisons between Starvation and Reconstitution (Fig. 6B). Again, it is difficult to interpret whether Fanmao's distinctiveness from the other two conditions is underlain by fascinating biology or technical batch effects. Without biological replicates, it remains challenging to disentangle the two.

As highlighted by the reviewer, our experimental design involves pooling multiple biological samples within a single treatment state before sequencing. We acknowledge the concern regarding the absence of distinct biological replicates and the potential impact of batch effects on result interpretation. While we recognize the merit of conducting multiple sequencing runs for a single treatment to provide genuine biological replicates, we contend that batch effects may not exert a strong influence on the observed patterns.

In addition, we applied a bootstrap sampling algorithm to assess whether the gene expression patterns within a cluster are more similar than those between clusters. This algorithm involves selecting a portion of cells per cluster and examining whether this subset remains distinguishable from other clusters. Our assumption was that if different samples exhibited distinct expression patterns due to batch effect, the co-assignment probabilities of a cluster would be very low. This expectation was not met in our data, as illustrated in Fig. S2. The lack of significantly low co-assignment probabilities within clusters suggests that batch effects may not exert a strong influence on our results.

Indeed, we acknowledge a noticeable shift in the expression patterns of certain cell types, such as the bacteriocyte. However, this is not universally applicable across all cell types. For instance, the UMAP figure in Fig. 6A illustrates a substantial overlap among basal membrane cell 2 from Fanmao, Starvation, and Reconstitution treatments, and the centroid distances between the three treatments are subtle, as depicted in Fig. 6B. This consistent pattern is also observed in DEPC, smooth muscle cells, and the food groove ciliary cells.

The reviewer also noted variations in the number of cells per treatment. Specifically, Fanmao sequencing yielded fewer than 10 thousand cells, whereas the other two treatments produced 2-3 times more cells after quality control (QC). It is highly probable that the technician loaded different quantities of cells into the machine for single-nucleus sequencing—a not uncommon occurrence in this methodology. While loading more cells may increase the likelihood of doublets, it is crucial to emphasize that this should not significantly impact the expression patterns post-QC. It's worth noting that overloading samples has been employed as a strategic approach to capture rare cell types, as discussed in a previous study (reference: 10.1126/science.aay0267).

The reviewer highlighted the discrepancy in cell survival rates during the 'doublet filtering' process, with 98% of Fanmao cells surviving compared to approximately 40% and 70% for the Starvation and Reconstitution conditions, respectively. It's important to clarify that the reported percentages reflect the survival of cells through a multi-step QC process employing various filtering strategies.

Post-doublet removal, we filtered out cells with <100 or >2500 genes and <100 or >6000 unique molecular identifiers (UMIs). Additionally, genes with <10 UMIs in each data matrix were excluded. The observed differences in survival rates for Starvation and Reconstitution cells can be attributed to the total volume of data generated in Illumina sequencing. Specifically, we sequenced approximately 91 GB of data for Fanmao, ~196 GB for Starvation, and ~249 GB for Reconstitution. As a result, the qualified data obtained for Starvation and Reconstitution conditions was only about twice that of Fanmao due to the limited data volume.

The reviewer also observed a divergence in the relative proportions of cells per cell type cluster in Fanmao compared to Reconstitution and Starvation, as depicted in Fig. S1. This discrepancy may hold true biological significance, presenting a potentially intriguing finding. However, our discussion on this pattern was rather brief, as we acknowledge that the observed differences could be influenced by the sample preparation process for dissection and digestion. It is crucial to consider that cutting a slightly different area during dissection may result in variations in the proportion of cells obtained. While we recognize the potential impact of this factor, we do not think that the sparsity of sampling alone could significantly affect the relative proportions of cells per cell type.

In conclusion, we acknowledge the reviewer's suggestion that sequencing multiple individual samples per treatment condition would have been ideal, rather than pooling them together. However, the homogenous distribution observed in UMAP and the consistent results obtained from bootstrap sampling suggest that the impact of batch effects on our analyses is likely not substantial. Additionally, based on our understanding, the smaller number of cells in the Fanmao sample should not have any significant effect on the resulting different proportion of cells or the expression patterns per each cluster.

Reviewer #3 (Public Review):

*Wang et al. explored the unique biology of the deep-sea mussel *Gigantidas platifrons* to understand the fundamental principles of animal-symbiont relationships. They used single-nucleus RNA sequencing and validation and visualization of many of the important cellular and molecular players that allow these organisms to survive in the deep sea. They demonstrate that a diversity of cell types that support the structure and function of the gill including bacteriocytes, specialized epithelial cells that host sulfur-oxidizing or methane-oxidizing symbionts as well as a suite of other cell types including supportive cells, ciliary, and smooth muscle cells. By performing experiments of transplanting mussels from one habitat which is rich in methane to methane-limited environments, the authors showed that starved mussels may consume endosymbionts versus in methane-rich environments upregulated genes involved in glutamate synthesis. These data add to the growing body of literature that organisms control their endosymbionts in response to environmental change.*

The conclusions of the data are well supported. The authors adapted a technique that would have been technically impossible in their field environment by preserving the tissue and then performing nuclear isolation after the fact. The use of single-nucleus sequencing opens the possibility of new cellular and molecular biology that is not possible to study in the field. Additionally, the in-situ data (both WISH and FISH) are high-quality and easy to interpret. The use of cell-type-specific markers along with a symbiont-specific probe was effective. Finally, the SEM and TEM were used convincingly for specific purposes in the case of showing the cilia that may support water movement.

We appreciate the valuable feedback provided by the reviewer on our study. It is encouraging to know that our work was found to be interesting and that they conducted a thorough evaluation of our research. We will take their constructive comments into account as we strive to develop and enhance our study. Thank the reviewer for all the input.

The one particular area for clarification and improvement surrounds the concept of a proliferative progenitor population within the gill. The authors imply that three types of proliferative cells within gills have long been known, but their study may be the first to recover molecular markers for these putative populations. The markers the authors present for gill posterior end budding zone cells (PEBZCs) and dorsal end proliferation cells (DEPCs) are not intuitively associated with cell proliferation and some additional

exploration of the data could be performed to strengthen the argument that these are indeed proliferative cells. The authors do utilize a trajectory analysis tool called Slingshot which they claim may suggest that PEBZCs could be the origin of all gill epithelial cells, however, one of the assumptions of this analysis is that differentiated cells are developed from the same precursor PEBZC population.

However, these conclusions do not detract from the overall significance of the work of identifying the relationship between symbionts and bacteriocytes and how these host bacteriocytes modulate their gene expression in response to environmental change. It will be interesting to see how similar or different these data are across animal phyla. For instance, the work of symbiosis in cnidarians may converge on similar principles or there may be independent ways in which organisms have been able to solve these problems.

We are grateful for the valuable comments and suggestions provided by the reviewer. All suggestions have been carefully considered, and the manuscript has been revised accordingly. We particularly value the reviewer's insights regarding the characterization of the *G. platifrons* gill proliferative cell populations. In a separate research endeavor, we have conducted experiments utilizing both cell division and cell proliferation markers on these proliferative cell populations. While these results are not incorporated into the current manuscript, we would be delighted to share our preliminary findings with the reviewer. Our preliminary results indicate that the proliferative cell populations exhibit positivity for cell proliferation markers and contain a significant number of mitotic cells..

Recommendations for the authors:

Reviewer #1 (Recommendations For The Authors):

Further experiments are needed to link the changes in transcriptomes of Bathymodioline mussels in the different environmental conditions to changes in their interactions with symbiotes. For example, quantifying the abundance and comparing the morphology of symbiotes between the environmental conditions would lend much support for shifting between milking and farming strategies. Without analyzing the symbiotes and comparing them across populations, it is difficult to comment on the mechanisms of interactions between symbiotes and the hosts. Without this analysis, this data is better suited towards comments about the general effect of environmental perturbation and stress on gene expression in these mussels.

We appreciate the reviewer's comments. We are also very curious about the symbiont responses, especially at the single-cell level. However, all the current commercial single-cell RNA-seq technologies are based on oligo dT priming for reverse transcription and barcoding. Therefore, the bacterial gene expression information is omitted from our dataset. Hopefully, with the development of technology, we could conduct an integrated analysis of both host and symbiont gene expression soon.

Additionally, clarification is needed on which types of symbiotes are being looked at. Are they MOX or SOX populations? Are they homogenous? What are the concentrations of sulfur at the sampled sites?

We thank you for your valuable comments and suggestions. *Gigantidas platifrons* harbors a MOX endosymbiont population characterized by a single 16S rRNA phylotype. We apologize for any confusion resulting from our previous wording. To clarify, we have revised lines 57-59 of our introduction

In the text and images, consider using standardized gene names and leaving out the genome coordinates. This would greatly help with readability. Also, be careful to properly

follow gene naming and formatting conventions (ie italicizing gene names and symbols).

We appreciate the reviewer's insightful comments. In model animals, gene nomenclature often stems from forward genetic approaches, such as the identification of loss-of-function mutants. These gene names, along with their protein products, typically correspond to unique genome coordinates. Conversely, in non-model invertebrates (e.g., *Gigantidas platifrons* of present study), gene prediction relies on a combination of bioinformatics methods, including de novo prediction, homolog-based prediction, and transcriptomics mapping. Subsequently, the genes are annotated by identifying their best homologs in well-characterized databases. Given that different genes may encode proteins with similar annotated functions, we chose to include both the gene ID (genome coordinates) and the gene name in our manuscript. This dual labeling approach ensures that our audience receives accurate and comprehensive information regarding gene identification and annotation.

Additionally, extending KEGG analysis to the atlas annotation section could help strengthen the confidence of annotations. For example, when identifying bacteriocyte populations, the functional categories of individual marker genes (lysosomal proteases, lysosomal traffic regulators, etc) are used to justify the annotation. Presenting KEGG support that these functional categories are upregulated in this population relative to others would help further support how you characterize this cluster by showing it's not just a few specific genes that are enriched in this cell group, but rather an overall functionality.

We appreciate the valuable suggestion provided by the reviewer. Indeed, incorporating KEGG analysis into the atlas annotation section could further enhance the confidence in our annotations. However, in our study, we encountered some limitations that impeded us from conducting a comprehensive KEGG enrichment analysis.

Firstly, the number of differentially expressed genes (DEGs) that we identified for certain cell populations was relatively small, making it challenging to meet the threshold required for meaningful KEGG enrichment analysis. For instance, among the 97 marker genes identified for the Bacteriocyte cluster, only two genes, Bpl_scaf_59648-4.5 (lysosomal alpha-glucosidase-like) and Bpl_scaf_52809-1.6 (lysosomal-trafficking regulator-like isoform X1), were identified as lysosomal genes. To generate reliable KEGG enrichments, a larger number of genes is typically required.

Secondly, single-nucleus sequencing, as employed in our study, tends to yield a relatively smaller number of genes per cell compared to bulk RNA sequencing. This limited gene yield can make it challenging to achieve sufficient gene representation for rigorous KEGG enrichment analysis.

Furthermore, many genes in the genome still lack comprehensive annotation, both in terms of KEGG and GO annotations. In our dataset, out of the 33,584 genes obtained through single-nuclei sequencing, 26,514 genes have NO KEGG annotation, and 25,087 genes have NO GO annotation. This lack of annotations further restricts the comprehensive application of KEGG analysis in our study.

The claim that VEPCs are symbiote free is not demonstrated. Additional double in situ are needed to show that markers of this cell type localize in regions free of symbiotes.

We appreciate your comments and suggestions. In Figure 5B, our results demonstrate that the bacteriocytes (green fluorescent signal) are distant from the VEPCs, which are located around the tip of the gill filaments (close to the food groove). We have revised our Figure 5B to make it clear.

Additionally, it does not seem like trajectory analysis is appropriate for these sampling conditions. Generally, to create trajectories confidently, more closely sampled time points are needed to sufficiently parse out the changes in expression. More justification is needed for the use of this type of analysis here and a discussion of the limitations should be mentioned, especially when discussing the hypotheses relating to PEBZCs, VEPCs, and DEPCs.

We greatly appreciate your thoughtful commentary. It is important to acknowledge that in the context of a developmental study, incorporating more closely spaced time points indeed holds great value. In our ongoing project investigating mouse development, for instance, we have implemented time points at 24-hour intervals. However, in the case of deep-sea adult animals, we hypothesized a slower transcriptional shift in such extreme environment, which led us to opt for a time interval of 3-7 days. Examining the differential expression profiles among the three treatments, we observed that most cell types exhibited minimal changes in their expression profiles. For the cell types strongly impacted by in situ transplantation, their expression profiles per cell type still exhibited highly overlap in the UMAP analysis (Figure 6a), thus enabling meaningful comparisons. Nevertheless, we recognize that our sampling strategy may not be flawless. Additionally, the challenging nature of conducting in situ transplantation in 1000-meter depths limited the number of sampling occasions available to us. We sincerely appreciate your input and understanding.

Finally, more detail should be added on the computational methods used in this paper. For example, the single-cell genomics analysis protocol should be expanded on so that readers unfamiliar with BD single-cell genomics handbooks could replicate the analysis. More detail is also needed on what criteria and cutoffs were used to calculate marker genes. Also, please be careful to cite the algorithms and software packages mentioned in the text.

Acknowledged, thank you for highlighting this. In essence, the workflow closely resembles that of the 10x Genomics workflow (despite the use of a different software, i.e., Cell Ranger). We better explain the workflow below, and also noting that this information may no longer be relevant for newer users of BD or individuals who are not acquainted with BD, given that the workflow underwent a complete overhaul in the summer of 2023.

References to lines

Line 32: typo "...uncovered unknown tissue heterogeny" should read "uncovering" or "and uncovered")

Overall abstract could include more detail of findings (ex: what are the "shifts in cell state" in line 36 that were observed)

We apologize for the mistakes, and have revised the manuscript accordingly.

Line 60: missing comma "...gill filament structure, but also"

We apologize for the mistakes, and have revised the manuscript accordingly.

Line 62-63: further discussion here, or in the relevant sections of the specific genes identified in the referenced bulk RNA-seq project could help strengthen confidence in annotation

We appreciate the comment, and have revised the manuscript accordingly.

Line 112: what bootstrapping strategy? Applied to what?

This is a bootstrap sampling algorithm to assess the robustness of each cell cluster developed in a recent biorxiv paper. (Singh, P. & Zhai, Y. Deciphering Hematopoiesis at single cell level through the lens of reduced dimensions. bioRxiv, 2022.2006.2007.495099 (2022). <https://doi.org/10.1101/2022.06.07.495099>)

Lines 127-129: What figures demonstrate the location of the inter lamina cells? Are there in situs that show this?

We apologize for any errors; the referencing of figures in the manuscript has been revised for clarity

Lines 185-190: does literature support these as markers of SMCs? Are they known smooth muscle markers in other systems?

We characterized the SMCs by the expression of LDL-associated protein, angiotensin-converting enzyme-like protein, and the "molecular spring" titin-like protein, all of which are commonly found in human vascular smooth muscle cells. Based on this analysis, we hypothesize that these cells belong to the smooth muscle cell category.

Line 201: What is meant by "regulatory roles"?

In this context, we are discussing the expression of genes encoding regulatory proteins, such as SOX transcription factors and secreted-frizzled proteins.

Line 211: which markers disappeared? What in situs show this?

We apologize for the mistakes, and have revised the manuscript accordingly.

Line 211: typo, "role" → "roll"

We apologize for the mistakes, and have revised the manuscript accordingly.

Line 214: what are these "hallmark genes"

We apologize for the mistakes, here we are referring to the genes listed in figure 4B. We have revised the manuscript accordingly.

Line 220: are there meristem-like cells in metazoans? If so, this would be preferable to a comparison with plants.

In this context, we are discussing the morphological characteristics of gill proliferative cell populations found in filibranch bivalves. These populations, namely PEPC, VEPC, and DEPC, consist of cells exhibiting morphological traits akin to those of plant cambial-zone meristem cells. These cells typically display small, round shapes with a high nucleus-to-plasma ratio. We acknowledge that while these terms are utilized in bivalve studies (citations below), they lack the robust support seen in model systems backed by molecular biology evidences. The present snRNA-seq data, however, may offer valuable cell markers for future comprehensive investigations.

Leibson, N. L. & Movchan, O. T. Cambial zones in gills of Bivalvia. Mar. Biol. 31, 175-180 (1975). <https://doi.org/10.1007/BF00391629>

Wentrup, C., Wendeborg, A., Schimak, M., Borowski, C. & Dubilier, N. Forever competent: deep-sea bivalves are colonized by their chemosynthetic symbionts throughout their lifetime. *Environ. Microbiol.* 16, 3699-3713 (2014). <https://doi.org/10.1111/1462-2920.12597>

Cannuel, R., Beninger, P. G., McCombie, H. & Boudry, P. Gill Development and its functional and evolutionary implications in the blue mussel *Mytilus edulis* (Bivalvia: Mytilidae). *Biol. Bull.* 217, 173-188 (2009). <https://doi.org/10.1086/BBLv217n2p173>

Line 335: what is slingshot trajectory analysis? Does this differ from the pseudotime analysis?

Slingshot is an algorithm that uses the principal graph of the cells to infer trajectories. It models trajectories as curves on the principal graph, capturing the progression and transitions between different cellular states.

Both Slingshot and pseudotime aim to infer cellular trajectories. Slingshot focuses on capturing branching patterns which is fully compatible with the graph generated using dimensionality reduction such as UMAP and PHATE, while pseudotime analysis aims to order cells along a continuous trajectory. It does not rely on dimensionality reduction graphs. We used both in the MS for different purposes.

Line 241: introduce FISH methodology earlier in the paper, when in situ images are first referenced

We appreciate the comment, and have revised the manuscript accordingly.

Line 246-249: can you quantify the decrease in signal or calculate the concentration of symbiotes in the cells? Was 5C imaged whole? This can impact the fluorescent intensity in tissues of different thicknesses.

We appreciate your comment. In Figure 5C, most of the typical gill filament region is visible (the ventral tip of the gill filament, and the mid part of the gill filament) except for the dorsal end. The gill filament of bathymodioline mussels exhibits a simple structure: a single layer of bacteriocytes grow on the basal membrane. Consequently, the gill slices have a fairly uniform thickness (with two layers of bacteriocytes and one layer of interlamina cells in between), minimizing any potential impact on fluorescent intensity. As of now, detailed quantification of intracellular symbionts may necessitate continuous TEM or ultra-resolution confocal sections to 3D reconstruct the bacteriocytes, which may exceed the scope of the current study. Therefore, fluorescent intensity remains the only method available to us for estimating bacterial density/distribution across the gill filament.

Line 249: What is meant by 'environmental gradient?'

Here we are refereeing the gases need for symbiont's chemosynthesis. We have revised the manuscript to make it clear.

Lines 255-256: Were the results shown in the TEM images previously known? Not clear what novel information is conveyed in images Fig 5 C and D

In the Fig 5 C and D, we've delivered a high-quality SEM TEM image of a typical bacteriocyte, showcasing its morphology and subcellular machinery with clarity. These electron microscopy images offer the audience a comprehensive introduction to the cellular function of bacteriocytes. Additionally, they serve as supportive evidence for the bacteriocytes' snRNA-seq data.

Line 295-296: Can you elaborate on what types of solute carrier genes have been shown to be involved with symbioses?

We appreciate the comment, and have revised the manuscript accordingly. The putative functions of the solute carriers could be found in Figure 5I.

Line 297-301: Which genes from the bulk RNA-seq study? Adding more detail and references in cluster annotation would help readers better understand the justifications.

We appreciate the comment, and have revised the manuscript accordingly.

Line 316 -322: Can you provide the values of the distances?

We also provide values in the main text, in addition to the Fig6b. We also provide a supplementary Table (Supplementary Table S19).

Line 328: What are the gene expression patterns?

We observed genes that are up- and down-regulated in Starvation and reconstitution.

Line 334-337: A visualization of the different expression levels of the specific genes in clusters between sites might be helpful to demonstrate the degree of difference between sites.

We have prepared a new supplementary file showing the different expression levels.

Line 337: Citation needed

We appreciate the comment. Here, we hypothesize the cellular responds based on the gene's function and their expression patterns.

Line 402-403: Cannot determine lineages from data presented. Need lineage tracing over time to determine this

We acknowledge the necessity of conducting lineage tracing over time to validate this hypothesis. Nonetheless, in practical terms, it is difficult to obtain samples for testing this. Perhaps, it is easier to use their shallow sea relatives to test this hypothesis. However, in practice, it is very difficult.

413-414: What are the "cell-type specific responses to environmental change"? It could be interesting to present these results in the "results and discussion" section

These results are shown in Supplementary Figure S8.

Line 419-424: Sampling details might go better earlier on in the paper, when the sampling scheme is introduced.

We appreciate the comments. Here, we are discussing the limitations of our current study, not sampling details.

Line 552: What type of sequencing? Paired end? How long?

We conducted 150bp paired-end sequencing.

556-563: More detail here would be useful to readers not familiar with the BD guide. Also be careful to cite the software used in analysis!

The provided guide and handbook elucidate the intricacies of gene name preparation, data alignment to the genome, and the generation of an expression matrix. It is worth mentioning that we relied upon outdated versions of the aforementioned resources during our data analysis phase, as they were the only ones accessible to us at the time. However, we have since become aware of a newer pipeline available this year, rendering the information presented here of limited significance to other researchers utilizing BD.

Many thanks for your kind reminding. We have now included a reference for STAR. All other software was cited accordingly. There are no scholarly papers or publications to refer to for the BD pipeline that we can cite.

Line 577-578: How was the number of clusters determined? What is meant by "manually combine the clusters?" If cells were clustered by hand, more detail on the method is needed, as well as direct discussion and justification in the body of the paper.

It would be more appropriate to emphasize the determination of cell types rather than clusters. The clusters were identified using a clustering function, as mentioned in the manuscript. It's important to note that the clustering function (in our case, the FindClusters function of Seurat) provides a general overview based on diffuse gene expression. Technically speaking, there is no guarantee that one cluster corresponds to a single cell type. Therefore, it is crucial to manually inspect the clustering results to assign clusters to the appropriate cell types. In some cases, multiple clusters may be assigned to the same cell type, while in other cases, a single cluster may need to be further subdivided into two or more cell types or sub-cell types, depending on the specific circumstances.

For studies conducted on model species such as humans or mice, highly and specifically expressed genes within each cluster can be compared to known marker genes of cell types mentioned in previous publications, which generally suffices for annotation purposes. However, in the case of non-model species like Bathymodioline mussels, there is often limited information available about marker genes, making it challenging to confidently assign clusters to specific cell types. In such situations, in situ hybridisation proves to be incredibly valuable. In our study, WISH was employed to visualise the expression and morphology of marker genes within clusters. When WISH revealed the expression of marker genes from a cluster in a specific type of cell, we classified that cluster as a genuine cell type. Moreover, if WISH demonstrated uniform expression of marker genes from different clusters in the same cell, we assigned both clusters to the same cell type.

We expanded the description of the strategy in the Method section.

Line 690-692: When slices were used, what part of the gill were they taken from?

We sectioned the gill around the mid part which could represent the mature bacteriocytes.

References to figures:

General

Please split the fluorescent images into different channels with an additional composite. It is difficult to see some of the expression patterns. It would also make it accessible to colorblind readers.

We appreciate the comments and suggestions from the reviewer. We have converted our figures to CMYK colour which will help the colorblind audiences to read our paper.

Please provide the number of replicates for each in situ and what proportion of those displayed the presented pattern.

We appreciate the reviewer's comments. We have explained in the material and methods part of the manuscript.

Figure 2.C' is a fantastic summary and really helps the non-mussel audience understand the results. Adding schematics like this to Figures 3-5 would be helpful as well.

We value the reviewer's comments. We propose that Figures 3K, 4C, and 5A-D could offer similar schematic explanations to assist the audience.

Figure 2:

Figures 2.C-F, 2.C', 2.H-J are not referenced in the text. Adding in discussions of them would help strengthen your discussions on the cluster annotation

We appreciate the reviewer's comments. We have revised the manuscript accordingly.

In 2.B. 6 genes are highlighted in red and said to be shown in in situs, but only 5 are shown.

We apologize for the mistake. We didn't include the result 20639-0.0 WISH in present study. We have changed the label to black.

Figure 3:

Fig 2C-E not mentioned.

We appreciate the reviewer's comments. We have revised the manuscript accordingly.

In 3.B 8 genes are highlighted in red and said to be shown in in situs. Only 6 are.

The result of the WISH were provided in Supplementary Figures S4 and S5.

Figure 3.K is not referenced in the legend.

We appreciate the comment, and have revised the manuscript accordingly.

Figure 4:

In Figure D, it might be helpful to indicate the growth direction.

We appreciate the comment, and have revised the manuscript accordingly by adding an arrow in panel D to indicate growth direction.

4F: A double in situ with the symbiote marker is needed to demonstrate the nucleolin-like positive cells are symbiote free.

We appreciate the comment. The symbiont free region could be found in Figure 5A.

Figure 5:

In 5.A, quantification of symbiote concentration would help support your conclusion that they are denser around the edges.

We appreciate the comment, as we mentioned above, detailed quantification of intracellular symbionts may necessitate continuous TEM or ultra-resolution confocal sections to 3D reconstruct the bacteriocytes, which may exceed the scope of the current study. Therefore, fluorescent intensity remains the only method available to us for estimating bacterial density/distribution across the gill filament.

In 5.D, the annotation is not clear. Adding arrows like in 5.C would be helpful.

We appreciate the comment, and have revised the manuscript accordingly.

A few genes in 5.F are not mentioned in the paper body when listing other genes. Mentioning them would help provide more support for your clustering.

We appreciate the comment, and have revised the manuscript accordingly.

Is 5.I meant to be color coded with the gene groups from 5.F? Color Coding the gene names, rather than organelles or cellular structures might portray this better and help visually strengthen the link between the diagram and your dot plot.

We appreciate the suggestions. We've experimented with color-coding the gene names, but some colors are less discernible against a white background.

Figure 6:

6.B Is there a better way to visualize this data? The color coding is confusing given the pairwise distances. Maybe heatmaps?

We attempted a heatmap, as shown in the figure below. However, all co-authors agree that a bar plot provides clearer visualization compared to the heatmap. We agree that the color scheme may be confusing because they use the same color as for individual treatment. So we change the colors.

Author response image 1.

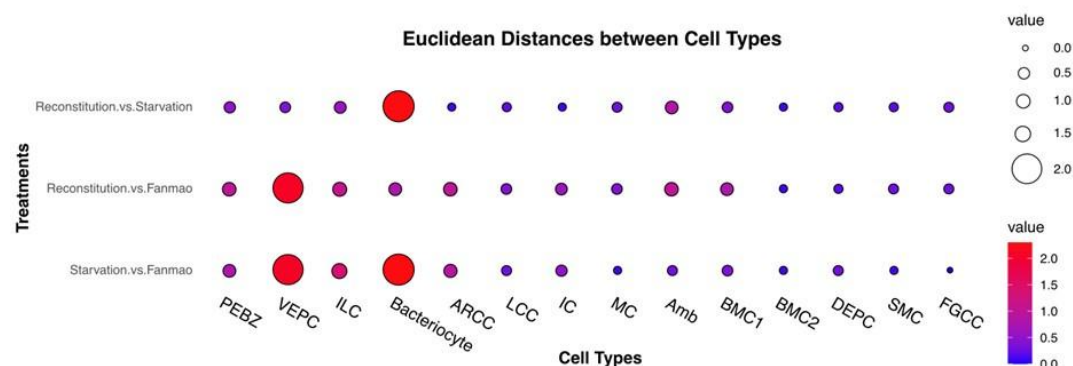


Figure 6.D: Why is the fanmao sample divided in the middle?

Fig6C show that single-cell trajectories include branches. The branches occur because cells execute alternative gene expression programs. Thus, in Fig 6D, we show changes for genes that are significantly branch dependent in both lineages at the same time. Specifically, in cluster 2, the genes are upregulated during starvation but downregulated during reconstitution. Conversely, genes in cluster 1 are downregulated during starvation but upregulated during reconstitution. It's of note that Fig 6D displays only a small subset of significantly branch-dependent genes.

Figure 6.D: Can you visualize the expression in the same format as in figures 2-5?

We appreciate the comments from the reviewer. As far as we know, this heatmap are the best format to demonstrate this type of gene expression profile.

Supplementary Figure S2:

Please provide a key for the cell type abbreviations

We appreciate the comment, and have added the abbreviations of cell types accordingly.

Supplementary Figures S4 and S5:

What part of the larger images are the subsetted image taken from?

We appreciate the comment, these images were taken from the ventral tip and mid of the gill slices, respectively. We have revised the figure legends to make it clear.

Supplemental Figure S7:

If clusters 1 and 2 show genes up and downregulated during starvation, what do clusters 4 and 3 represent?

Cluster 1: Genes that are obviously upregulated during Starvation, and downregulated during reconstitution; luster4 : genes are downregulated during reconstitution but not obviously upregulated during Starvation.

Cluster 2 show genes upregulated during reconstitution, and cluster 3 obviously downregulated during Starvation.

Author response table 1.

	<i>Fanmao -> Starvation</i>	<i>Fanmao -> reconstitution</i>
<i>Cluster 1</i>	<i>upregulate</i>	<i>downregulated</i>
<i>Cluster 2</i>	-	<i>upregulate</i>
<i>Cluster 3</i>	<i>downregulated</i>	-
<i>Cluster 4</i>	-	<i>downregulated</i>

Supplemental Figure S8:

This is a really interesting figure that I think shows some of the results really well! Maybe consider moving it to the main figures of the paper?

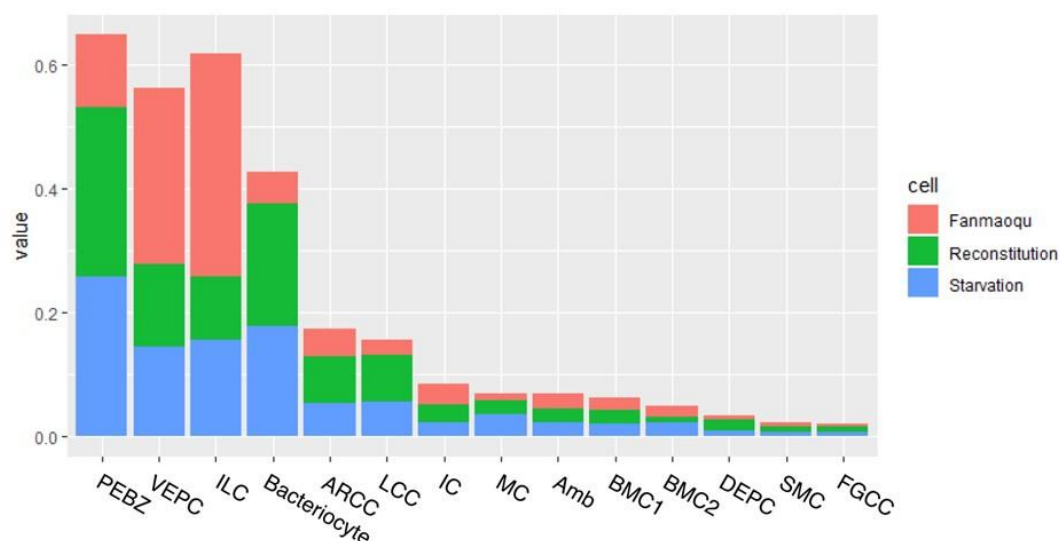
We appreciate the comments and suggestions. We concur with the reviewer on the significance of the results presented. However, consider the length of this manuscript, we have prioritized the inclusion of the most pertinent information in the main figures. Supplementary materials containing additional figures and details on the genes involved in these pathways are provided for interested readers.

Supplemental Figure S11:

Switching the axes might make this image easier for the reader to interpret. Additionally, calculating the normalized contribution of each sample to each cluster could help quantify the extent to which bacteriocytes are reduced when starving.

Thank you for the insightful suggestion, which we have implemented as detailed below. We acknowledge the importance of understanding the changes in bacteriocyte proportions across different treatments. However, it's crucial to note that the percentage of cells per treatment is highly influenced by factors such as the location of digestion and sequencing, as previously mentioned.

Author response image 2.



Reviewer #2 (Recommendations For The Authors):

The following are minor recommendations for the text and figures that may help with clarity:

Fig. 3K: This figure describes water flow induced by different ciliary cells. It is not clear what the color of the arrows corresponds to, as they do not match the UMAP (i.e. the red arrow) and this is not indicated in the legend. Are these colours meant to indicate the different ciliary cell types? If so it would be helpful to include this in the legend.

We appreciate the reviewer's comments and suggestions. The arrows indicate the water flow that might be agitated by the certain types of cilium. We have revised our figure and figure legends to make it clear.

Line 369: The incorrect gene identifier is given for the mitochondrial trifunctional enzyme. This gene identifier is identical to the one given in line 366, which describes long-chain-fatty-acid-ligase ACSBG2-like (Bpl_scaf_28862-1.5).

We appreciate the reviewer's comments and suggestions. We have revised our manuscript accordingly.

Line 554: The Bioproject accession number (PRJNA779258) does not appear to lead to an existing page in any database.

We appreciate the reviewer's comments and suggestions. We have released this Bioproject to the public.

Line 597-598: it would be helpful to know the specific number of cells that the three sample types were downsampled to, and the number of cells remaining in each cluster, as this can affect the statistical interpretation of differential expression analyses.

The number of cells per cluster in our analysis ranged from 766 to 14633. To mitigate potential bias introduced by varying cell numbers, we implemented downsampling,

restricting the number of cells per cluster to no more than 3500. This was done to ensure that the differences between clusters remained less than 5 times. We experimented with several downsampling strategies, exploring cell limits of 4500 and 2500, and consistently observed similar patterns across these variations.

Data and code availability:

The supplementary tables and supplementary data S1 appear to be the final output of the differential expression analyses. Including the raw data (e.g. reads) and/or intermediate data objects (e.g. count matrices, R objects), in addition to the code used to perform the analyses, may be very helpful for replication and downstream use of this dataset. As mentioned above, the Bioproject accession number appears to be incorrect.

We appreciate the reviewer's comments and suggestions. Regarding our sequencing data, we have deposited all relevant information with the National Center for Biotechnology Information (NCBI) under Bioproject PRJNA779258. Additionally, we have requested the release of the Bioproject. Furthermore, as part of this round of revision, we have included the count matrices for reference.

Reviewer #3 (Recommendations For The Authors):

As noted in the public review, my only major concerns are around the treatment of progenitor cell populations. I am sympathetic to the challenges of these experiments but suggest a few possible avenues to the authors.

*First, there could be some demonstration that these cells in *G. platifrons* are indeed proliferative, using EdU incorporation labeling or a conserved epitope such as the phosphorylation of serine 10 in histone 3. It appears in *Mytilus galloprovincialis* that proliferating cell nuclear antigen (PCNA) and phospho-histone H3 have previously been used as good markers for proliferative cells (Maiorova and Odintsova 2016). The use of any of these markers along with the cell type markers the authors recover for PEBZCs for example would greatly strengthen the argument that these are proliferative cells.*

If performing these experiments would not be currently possible, the authors could use some computation approaches to strengthen their arguments. Based on conserved cell cycle markers and the use of Cell-Cycle feature analysis in Seurat could the authors provide evidence that these progenitors occupy the G2/M phase at a greater percentage than other cells? Other than the physical position of the cells is there much that suggests that these are proliferative? While I am more convinced by markers in VEPCs the markers for PEBZCs and DEPCs are not particularly compelling.

While I do not think the major findings of the paper hinge on this, comments such as "the PBEZCs gave rise to new bacteriocytes that allowed symbiont colonization" should be taken with care. It is not clear that the PBEZCs are proliferative and there does not seem to be any direct evidence that PBEZCs (or DEPCs or VEPCs for that manner) are the progenitor cells through any sort of labeling or co-expression studies.

We appreciate the comments and suggestions from the reviewer. We have considered all the suggestions and have revised the manuscript accordingly. We especially appreciate the reviewer's suggestions about the characterisations of the *G. platifrons* gill proliferative cell populations. In a separate research project, we have tested both cell division and cell proliferation markers on the proliferation cell populations. Though we are not able to include these results in the current manuscript, we are happy to share our preliminary results with the reviewer. Our results demonstrate the proliferative cell populations, particularly the VEPCs, are cell proliferation marker positive, and contains high amount of mitotic cells.

Author response image 3.



Finally, there is a body of literature that has examined cell proliferation and zones of proliferation in mussels (such as Piquet, B., Lallier, F.H., André, C. et al. Regionalized cell proliferation in the symbiont-bearing gill of the hydrothermal vent mussel *Bathymodiolus azoricus*. *Symbiosis* 2020) or other organisms (such as Bird, A. M., von Dassow, G., & Maslakova, S. A. How the pilidium larva grows. *EvoDevo*. 2014) that could be discussed.

We appreciate the comments and suggestions from the reviewer. We have considered all the suggestions and have revised the manuscript accordingly (line 226-229).

Minor comments also include:

Consider changing the orientation of diagrams in Figure 2C' in relationship to Figure 2C and 2D-K.

We appreciate the comments and suggestions from the reviewer. The Figure 2 has been reorganized.

For the diagram in Figure 3K, please clarify if the arrows drawn for the direction of inter lamina water flow is based on gene expression, SEM, or some previous study.

We are grateful for the reviewer's valuable feedback and suggestions. The arrows in the figure indicate the direction of water flow that could be affected by specific types of cilium. Our prediction is based on both gene expression and SEM results. To further clarify this point, we have revised the figure legend of Fig. 3.

Please include a label for the clusters in Figure 5E for consistency.

We have revised our Figure 5E to keep our figures consistent.

Please include a note in the Materials and Methods for Monocle analysis in Figure 6.

We conducted Monocle analyses using Monocle2 and Monocle 3 in R environment. We have revised our material and methods with further information of Figure 6.

In Supplement 2, the first column is labeled PEBC while the first row is labeled PEBZ versus all other rows and columns have corresponding names. I am guessing this is a typo and not different clusters?

We appreciate the great effort of the reviewer in reviewing our manuscript. We have corrected the typo in the revised version.